Histology Lecture

Hepatobiliary system and pancreas

Dr. Radhakrishna
Shetty

MANIPAL
UNIVERSITY COLLEGE
MALAYSIA

Formerly known as MELAKA-MANIPAL MEDICAL COLLEGE



#inspire

Learning Objectives

At the end of the lecture, students should be able to:

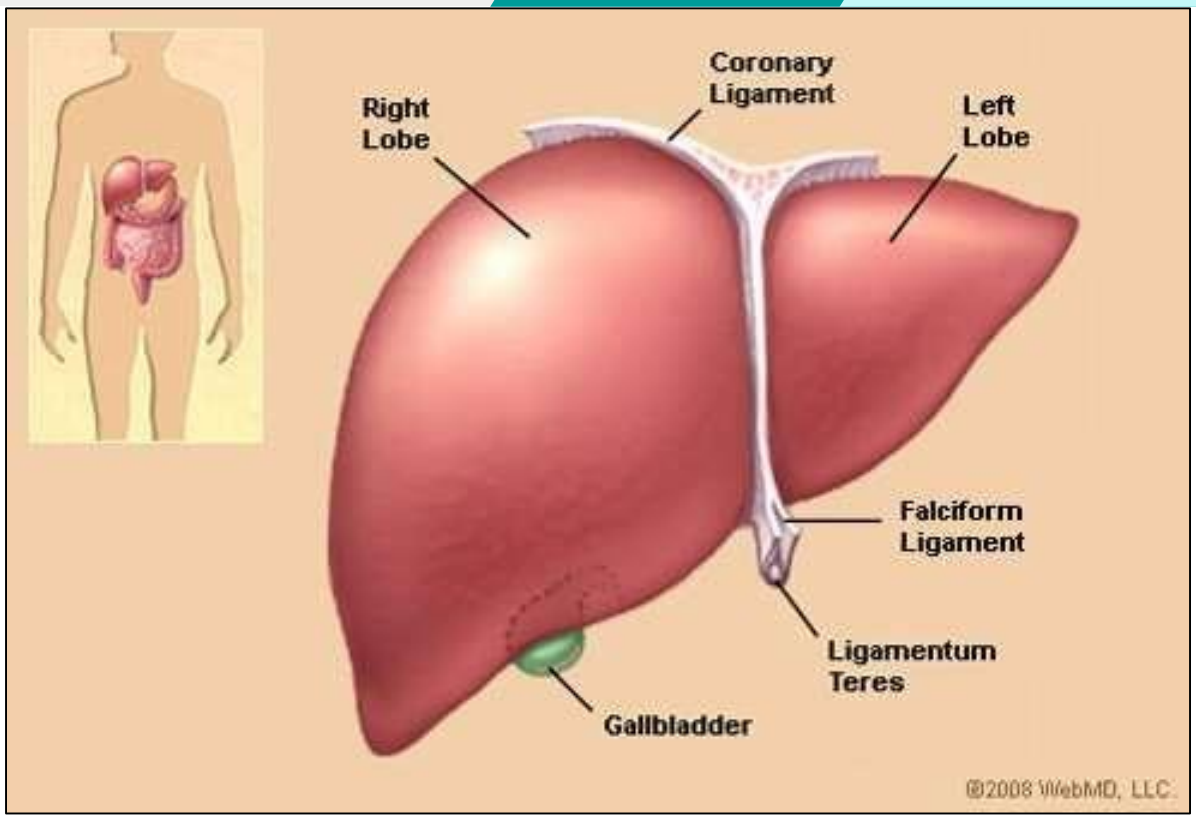
- Liver
  - Explain and illustrate the microscopic structure of liver
- Gall bladder
  - Explain and illustrate the microscopic structure of gall bladder
- Pancreas
  - Explain and illustrate the microscopic structure of pancreas

Liver

- Largest gland in the body.

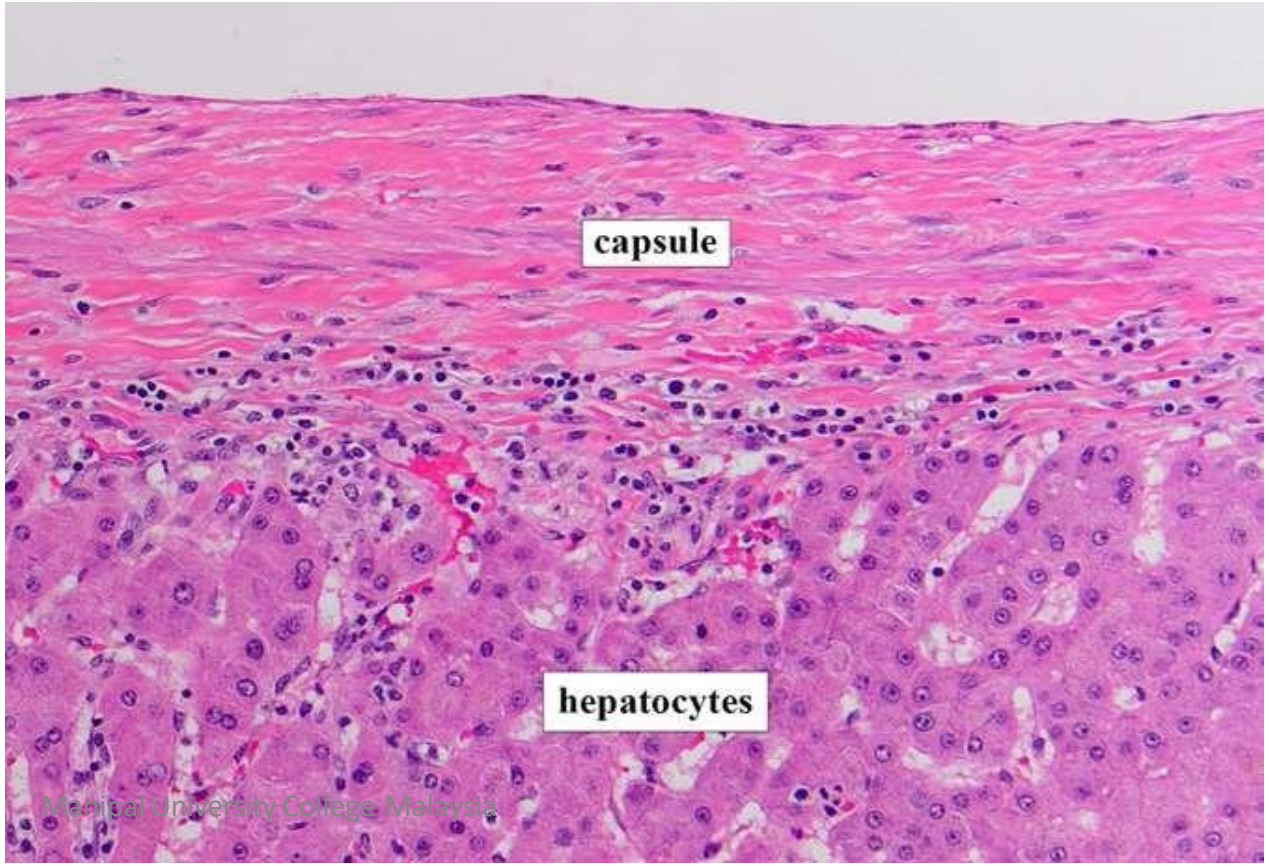
- Consists of:
  - ✓ a complex network of epithelial cells (hepatocytes)
  - ✓ supportive connective tissue
  - ✓ numerous blood vessels

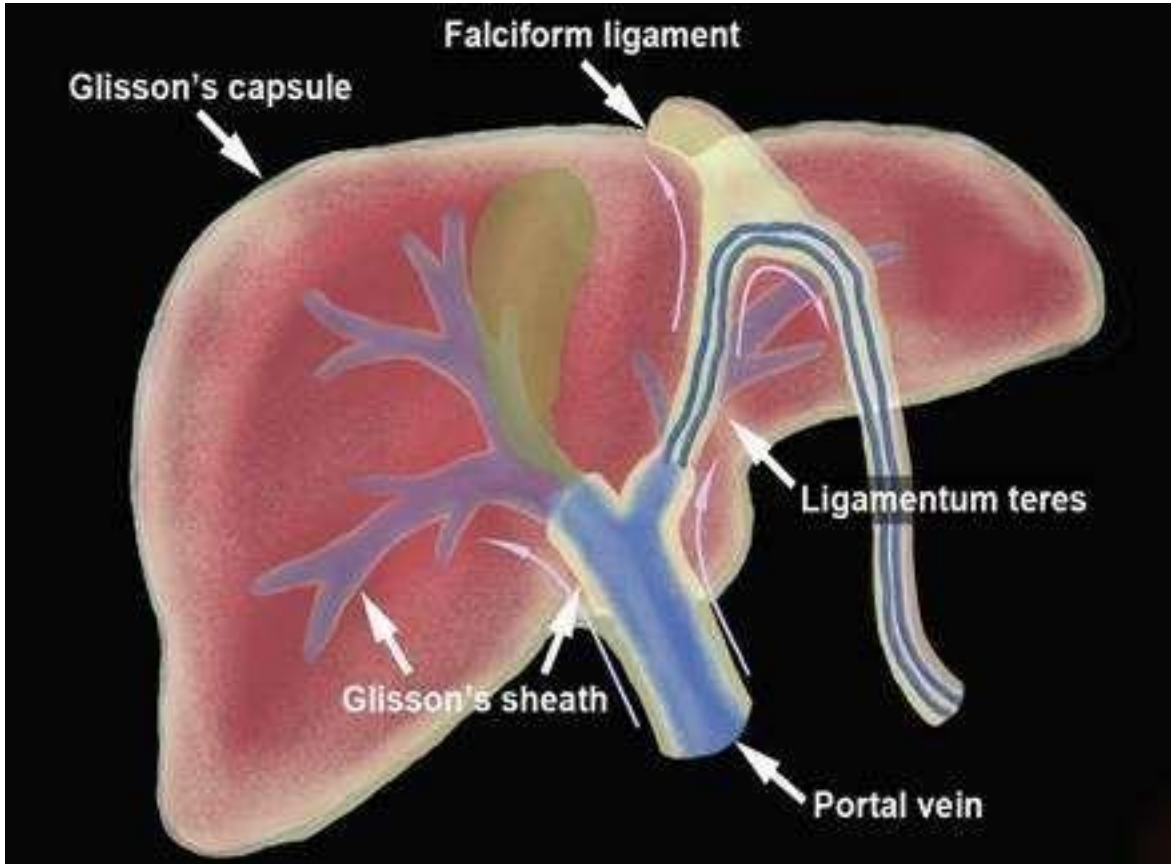
- Performs both endocrine and exocrine functions by the same cell.



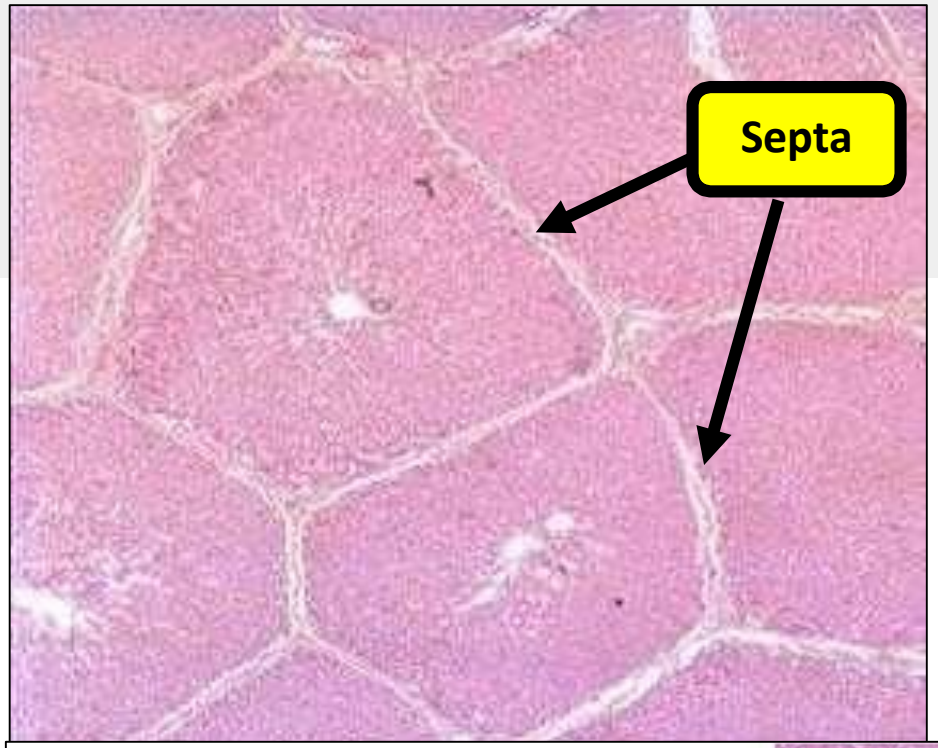
Structure

- Invested by a fibrous connective tissue capsule (**Glisson's capsule**).

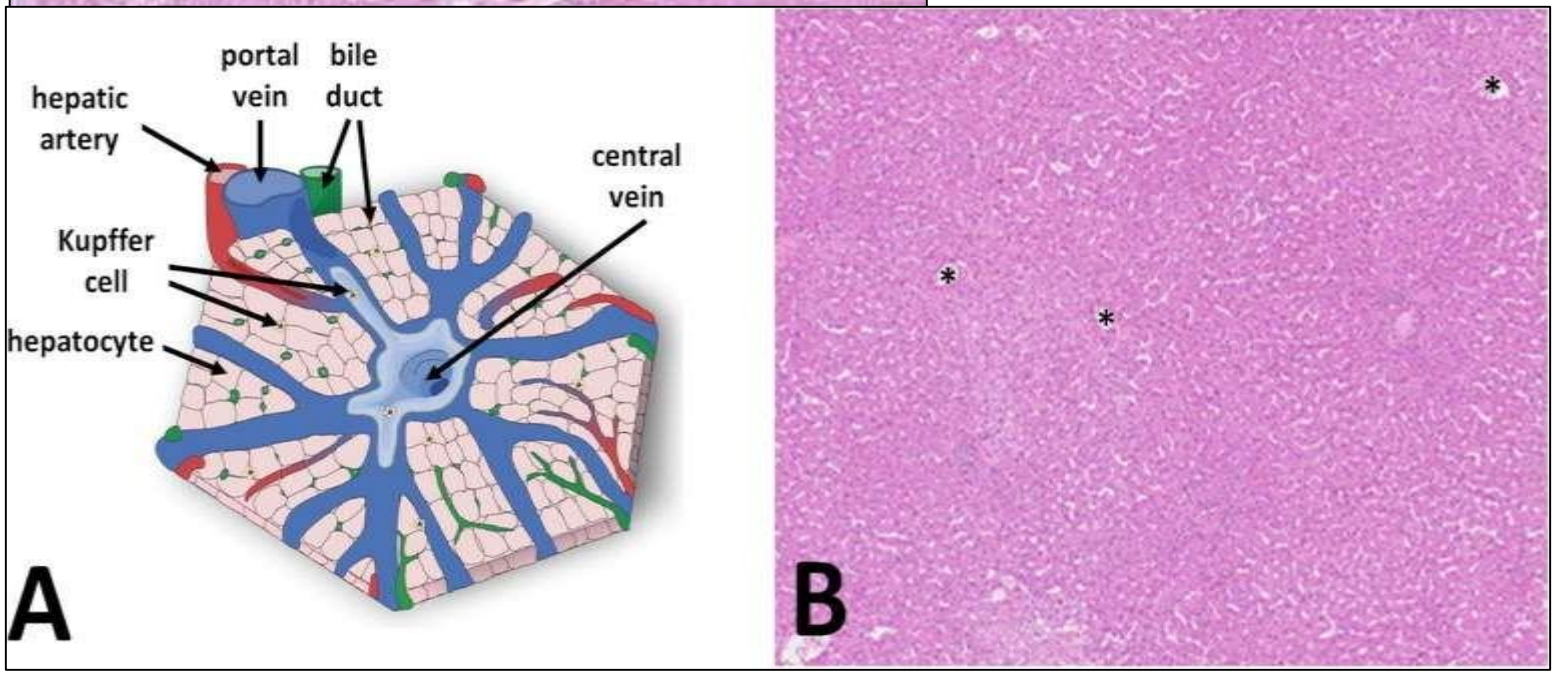




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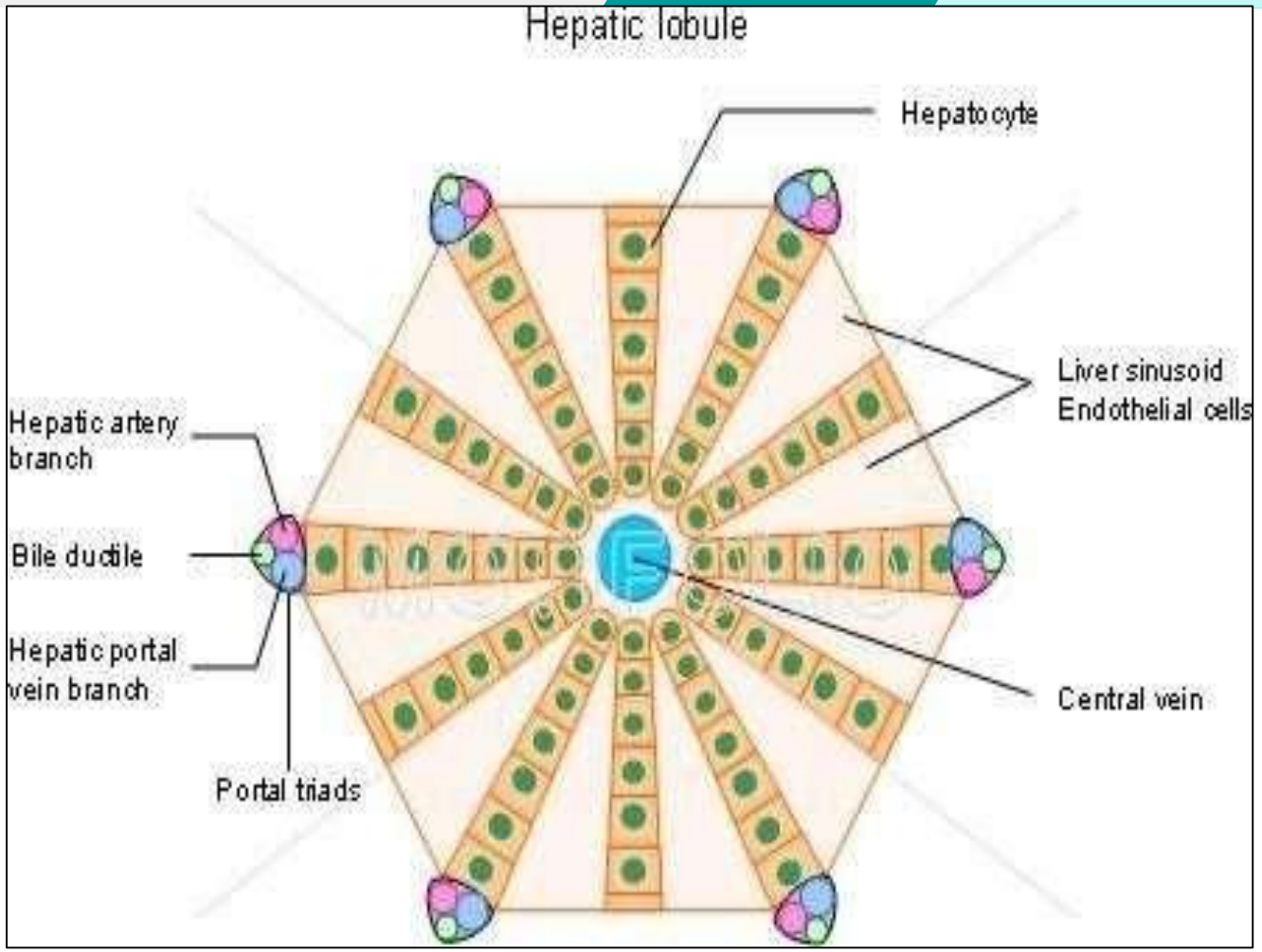
- The hepatic parenchyma is subdivided into small lobules called hepatic lobule by the **fibrous septa**.



- The parenchyma consists of **hepatocytes** and **bile ducts**.

Hepatic lobule

- Anatomical unit of the liver.
- Hexagonal/ Polygonal shape.
- Axis of lobule is **central vein.**
- Edges/corners of lobule are connective tissue and **portal triad.**

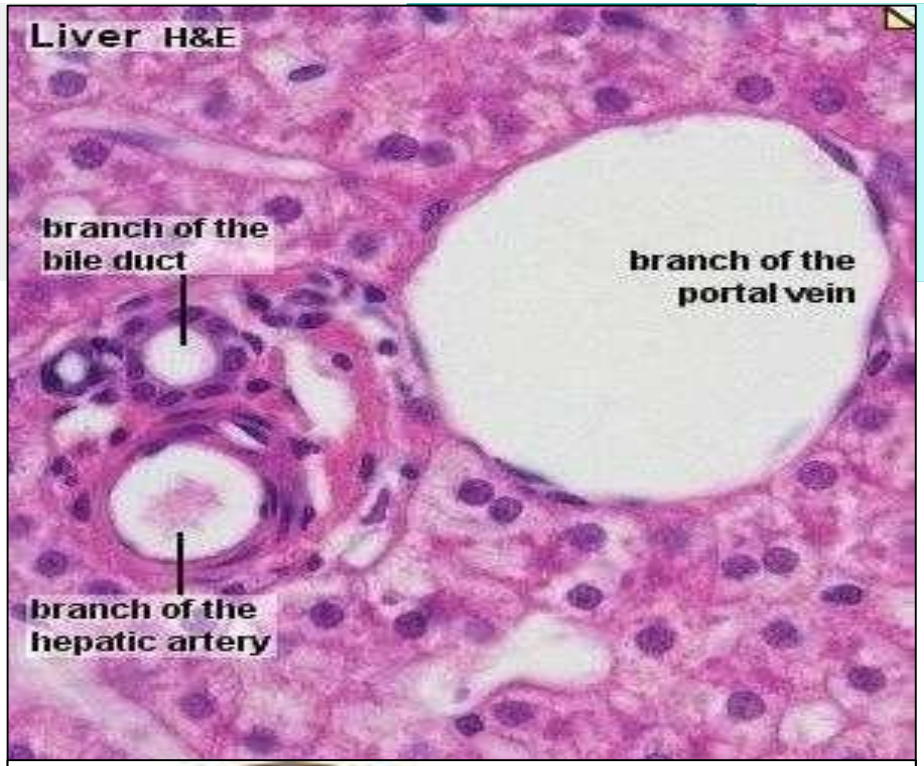


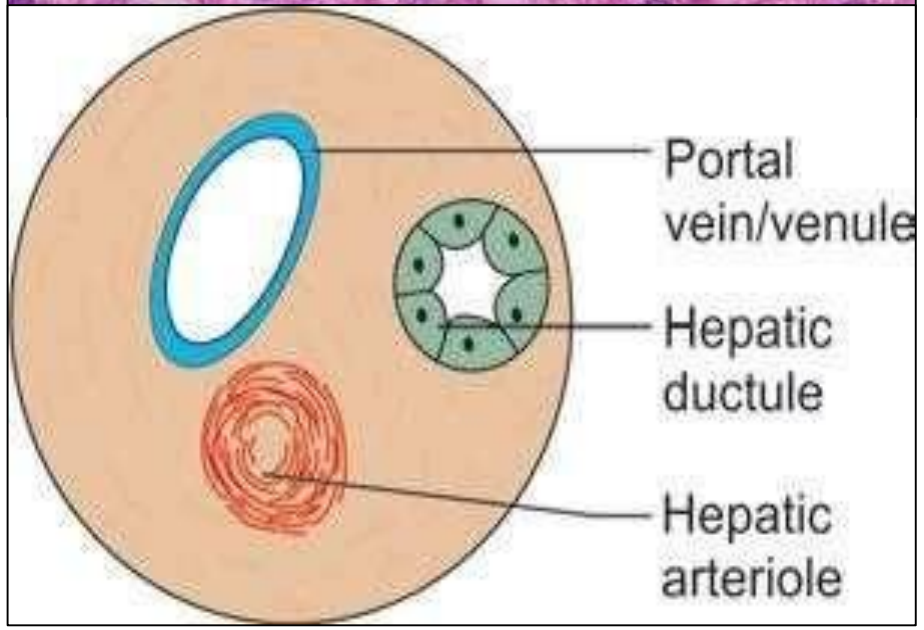
Portal Triad

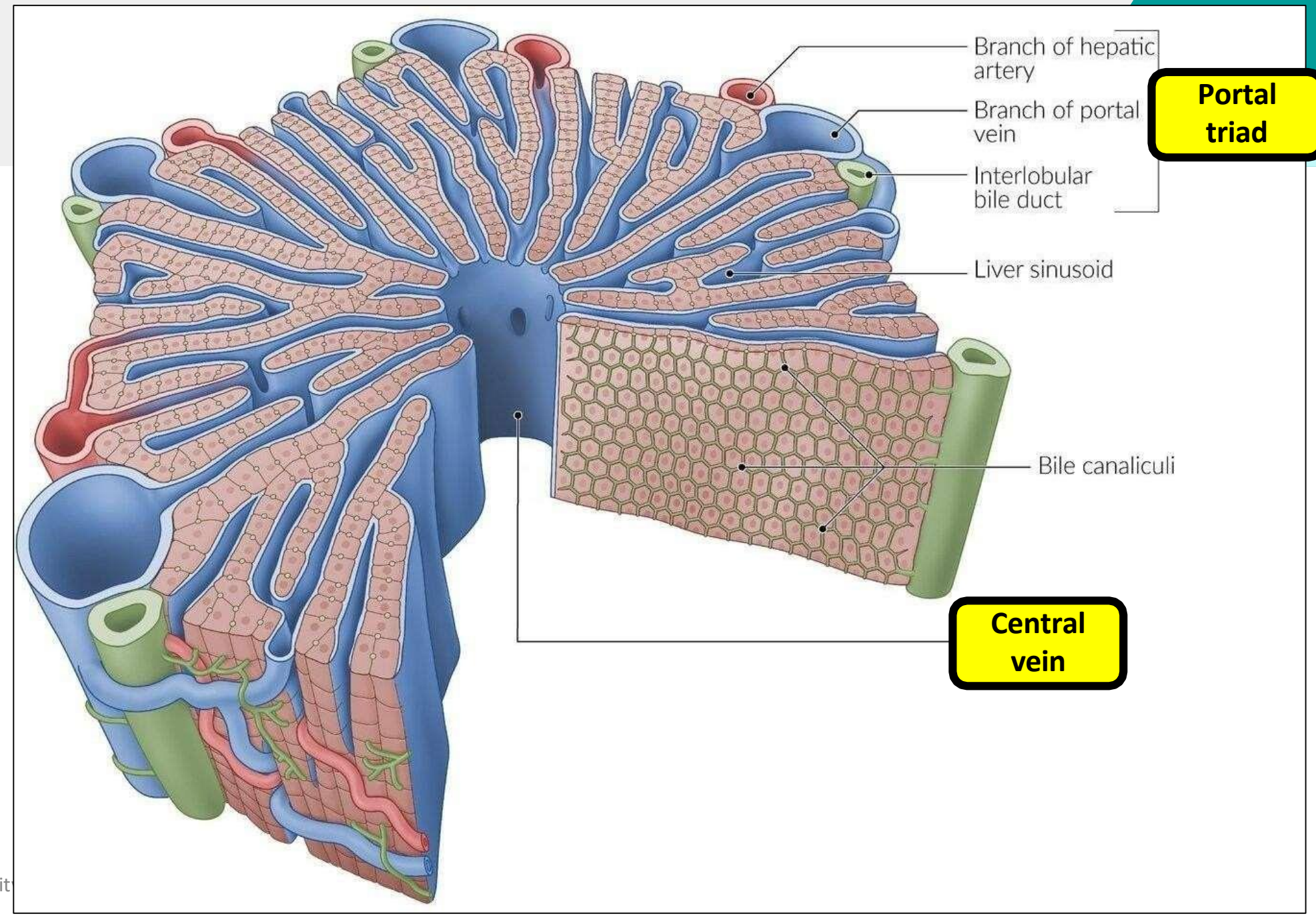
- Contents:
  - ✓ **Branch of hepatic artery**
  - ✓ **Portal vein (Largest)**
  - ✓ **Bile duct (Smallest)**

- Hepatic artery and portal vein are lined by endothelium.

- Bile duct is lined by cuboidal epithelium.

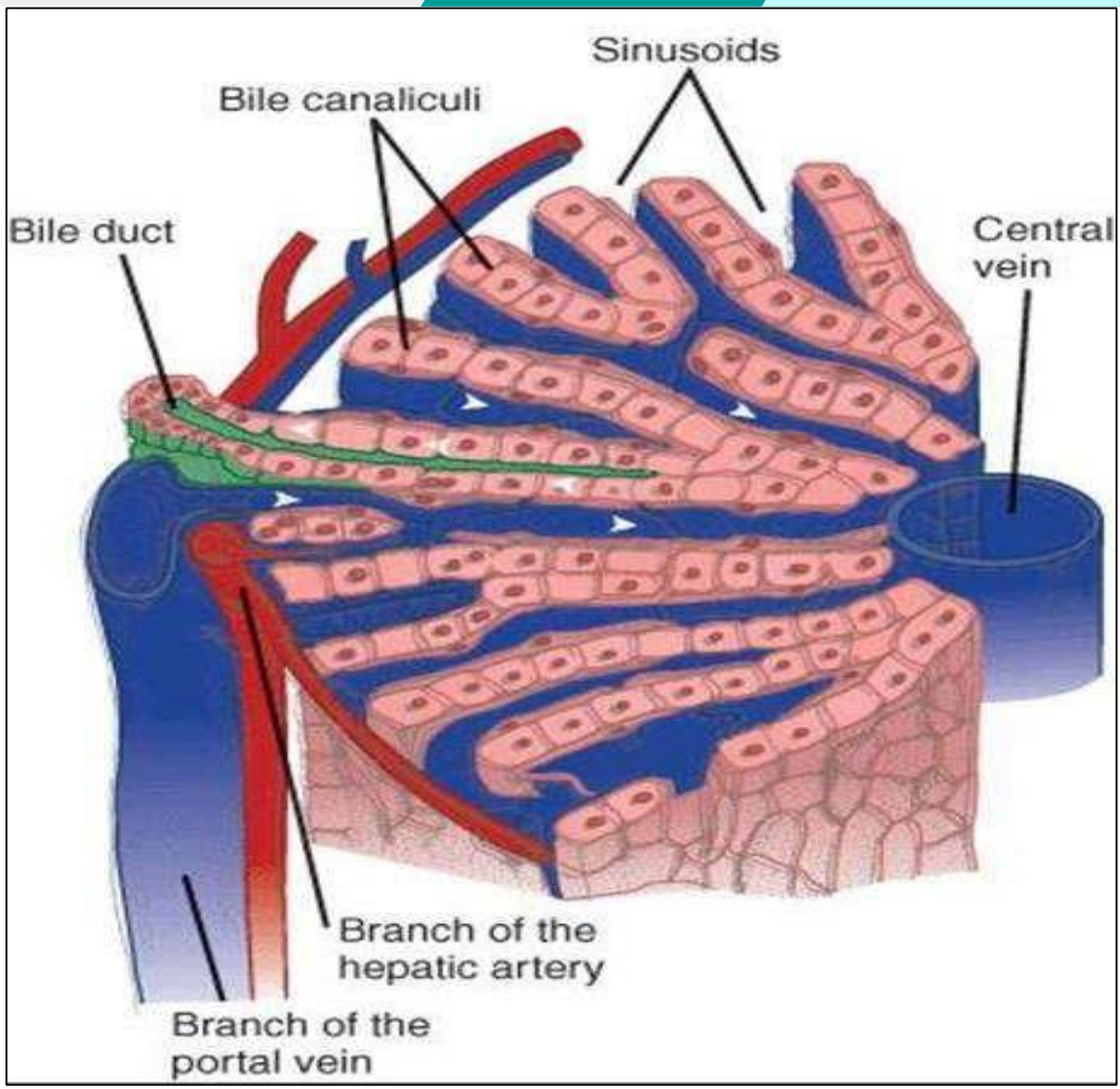


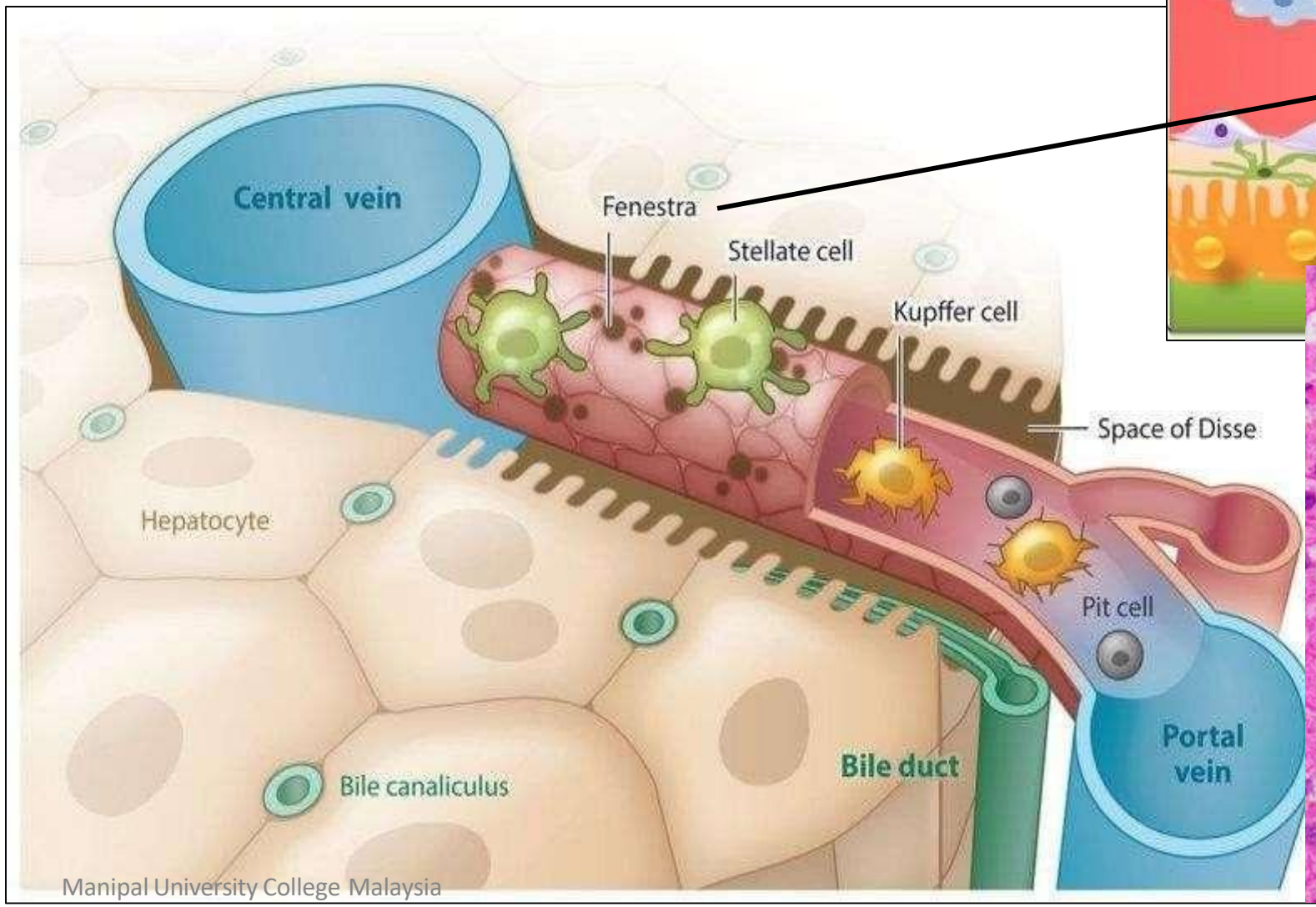


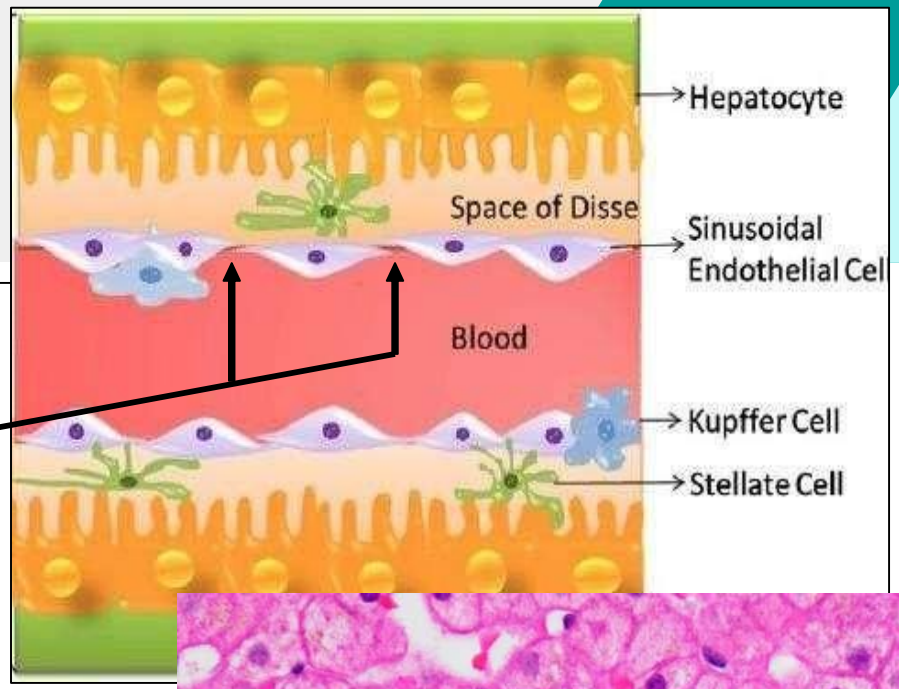

Branch of hepatic artery
Branch of portal vein
Interlobular bile duct
Portal triad
Liver sinusoid
Bile canaliculi
Central vein

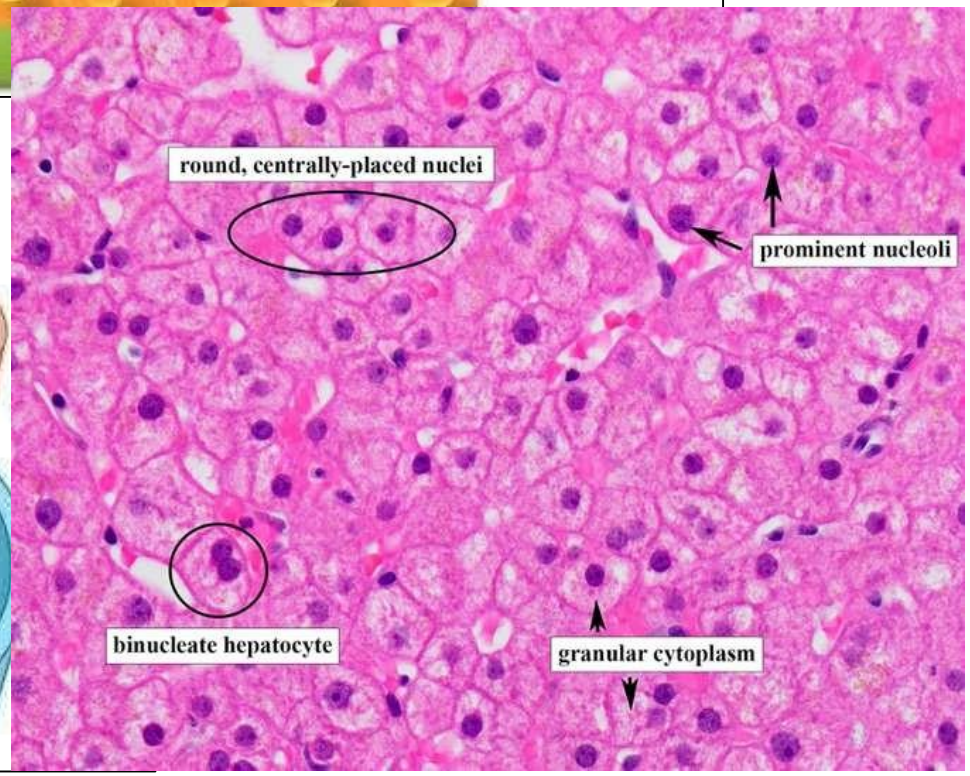
Hepatocytes

- Large polyhedral cell with rounded nucleus and eosinophilic cytoplasm.
- Grouped in plates (mostly one cell thick).
- 'Sinusoids' are located in between the cellular plates.
- 'Bile canaliculi' are present in between the hepatocytes which drain toward the periphery into bile duct in the portal triad.



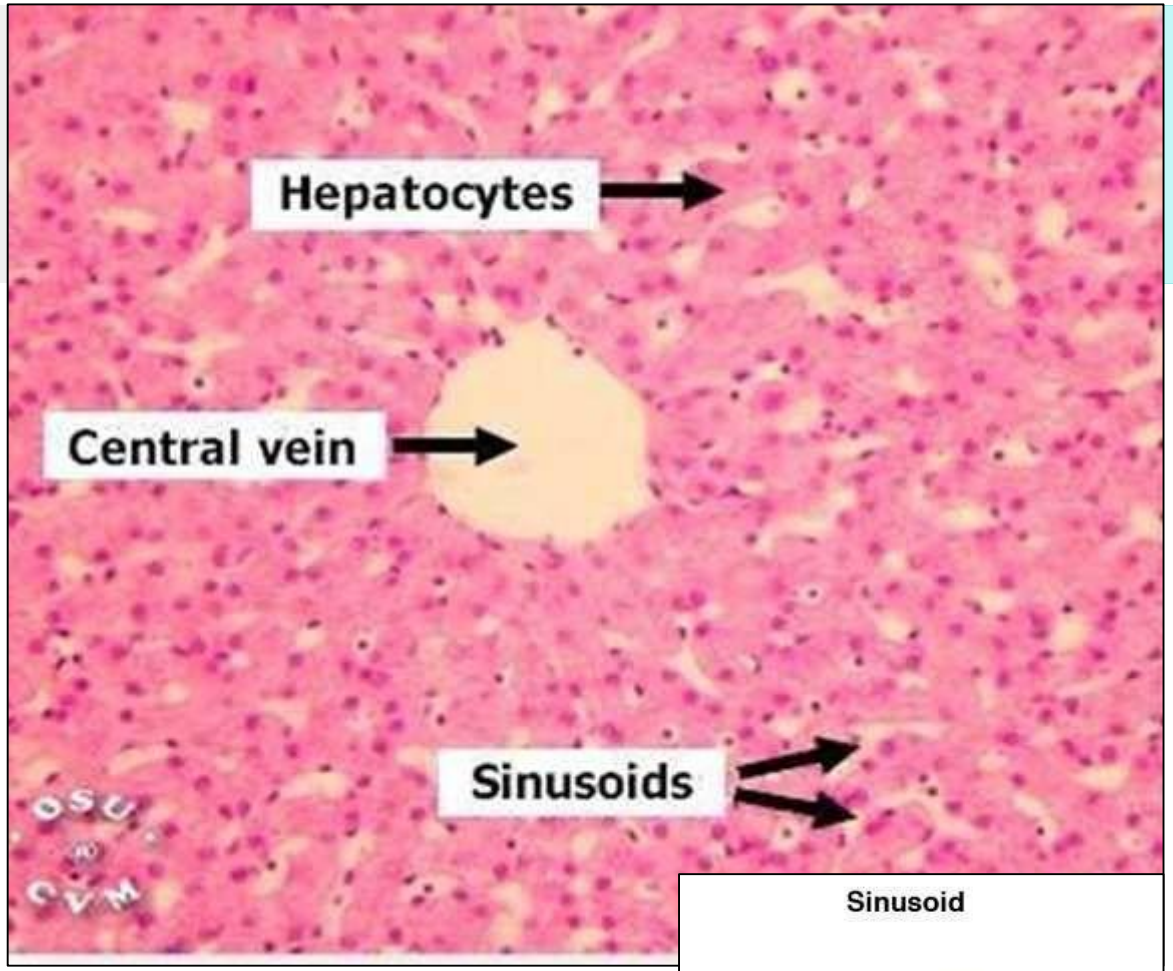

Central vein
Fenestra
Stellate cell
Kupffer cell
Space of Disse
Hepatocyte
Pit cell
Portal vein
Bile duct
Bile canaliculus

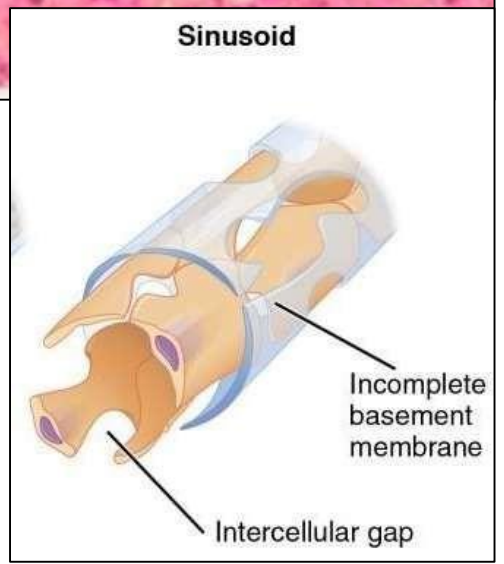

Hepatocyte
Space of Disse
Sinusoidal Endothelial Cel
Blood
Kupffer Cell
Stellate Cell

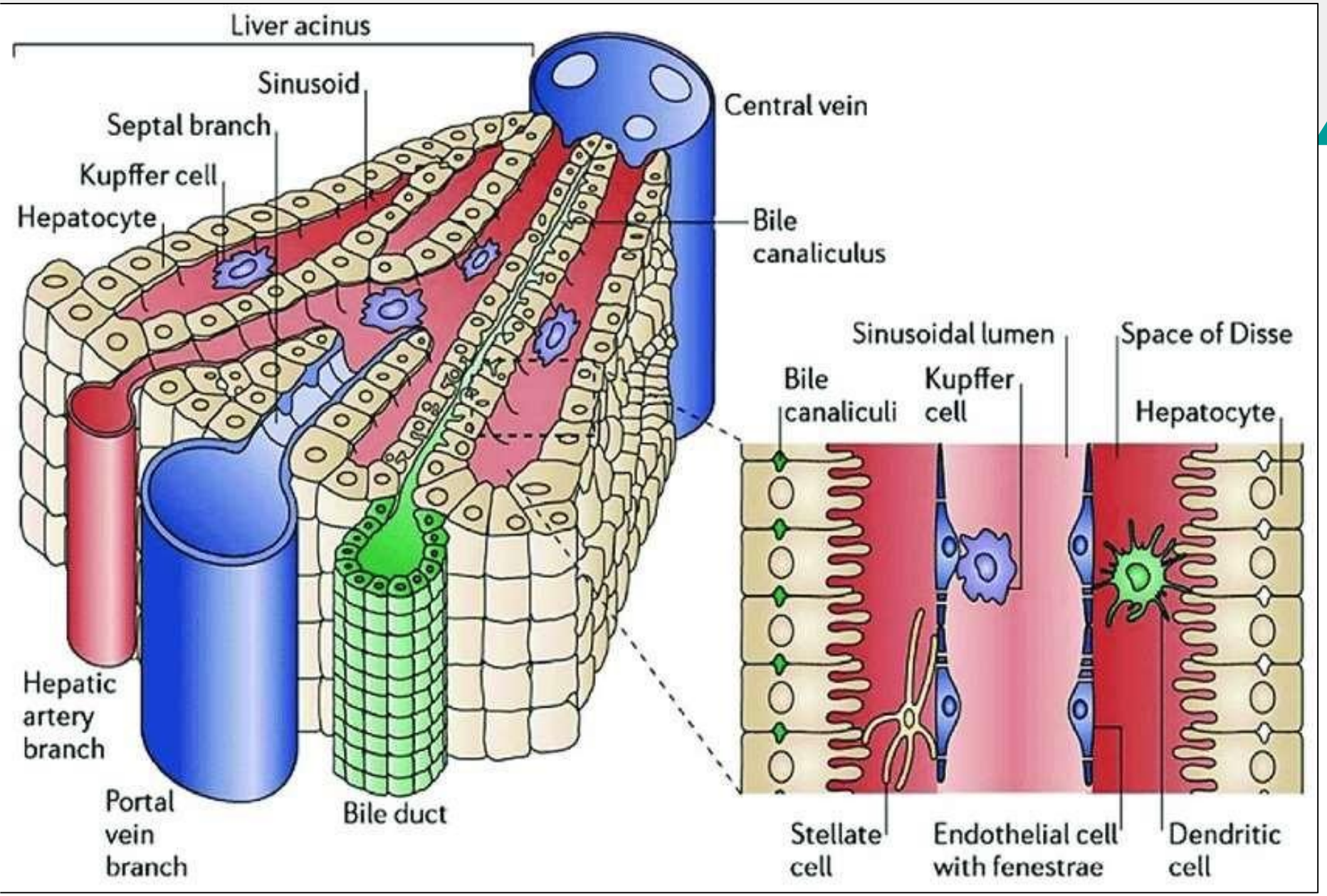

round, centrally-placed nuclei
prominent nucleoli
binucleate hepatocyte
granular cytoplasm

Liver sinusoids

- Occupy the space between cellular plates.
- Wall is composed of one discontinuous layer of endothelial cells.
- Contain phagocytic cells called **'Kupffer cells'.**





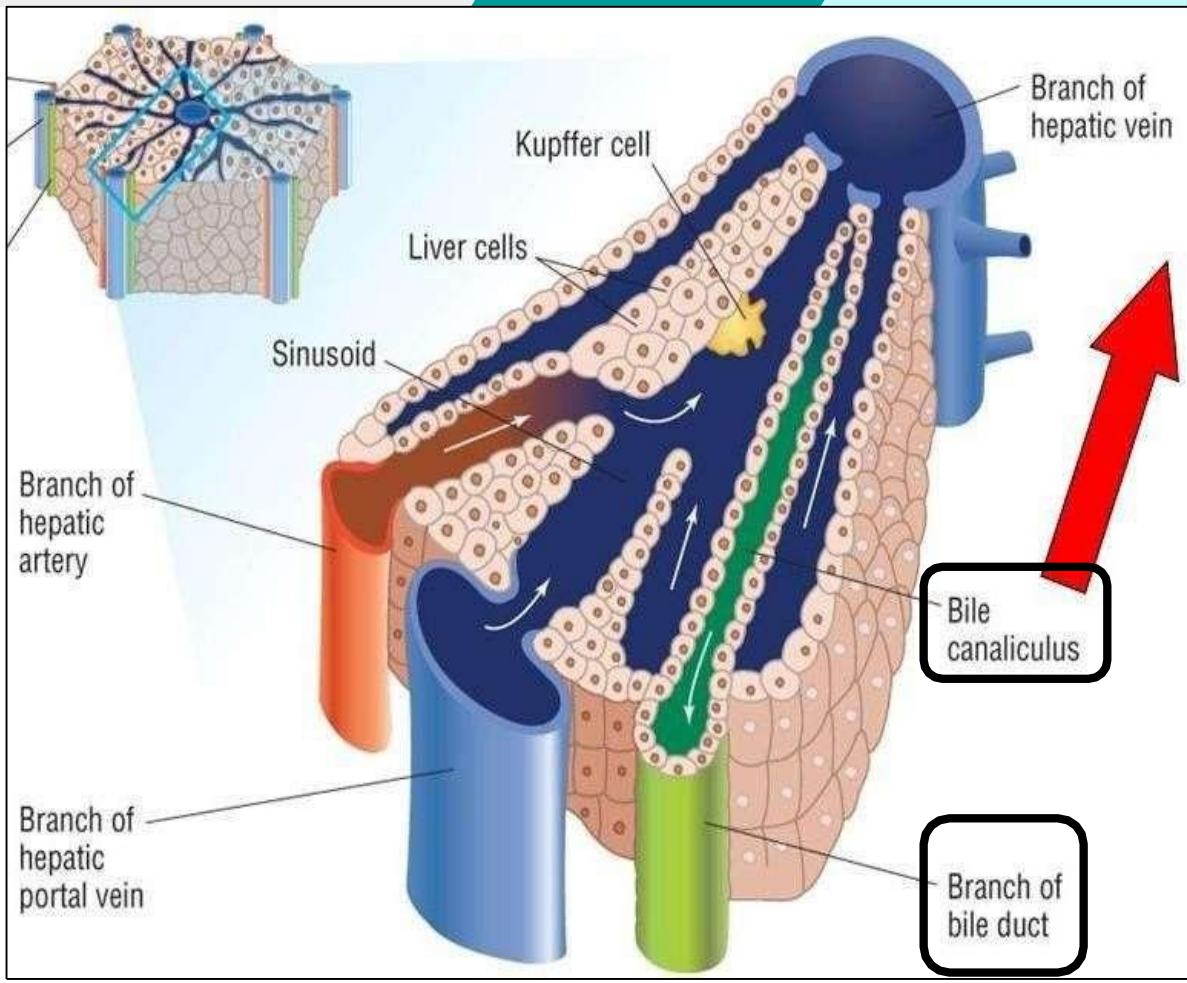


‘Space of Disse’

- ✓ **Subendothelial space** between endothelium and hepatocytes
- ✓ Containing **‘Stellate cells'.**

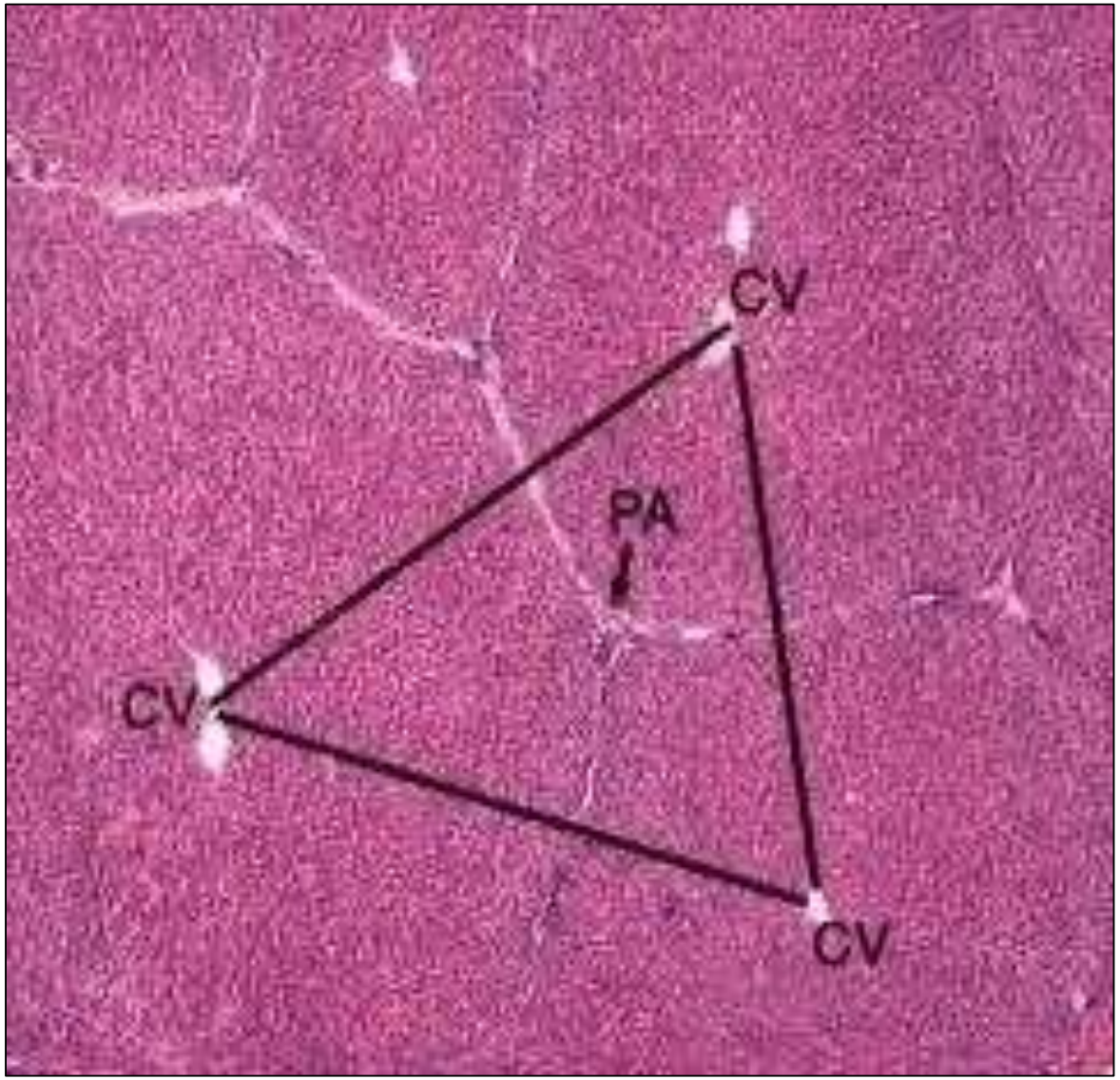
Bile duct system

- First portion of the bile duct system is bile canaliculi, tubular space between adjacent hepatocytes.
- They drain the bile produced by hepatocytes towards the periphery into the bile duct in the portal triad. These ducts are lined by cuboidal epithelium.
- As the size of the duct increase, the epithelium becomes columnar. They gradually enlarge and fuse forming the right and left hepatic ducts.

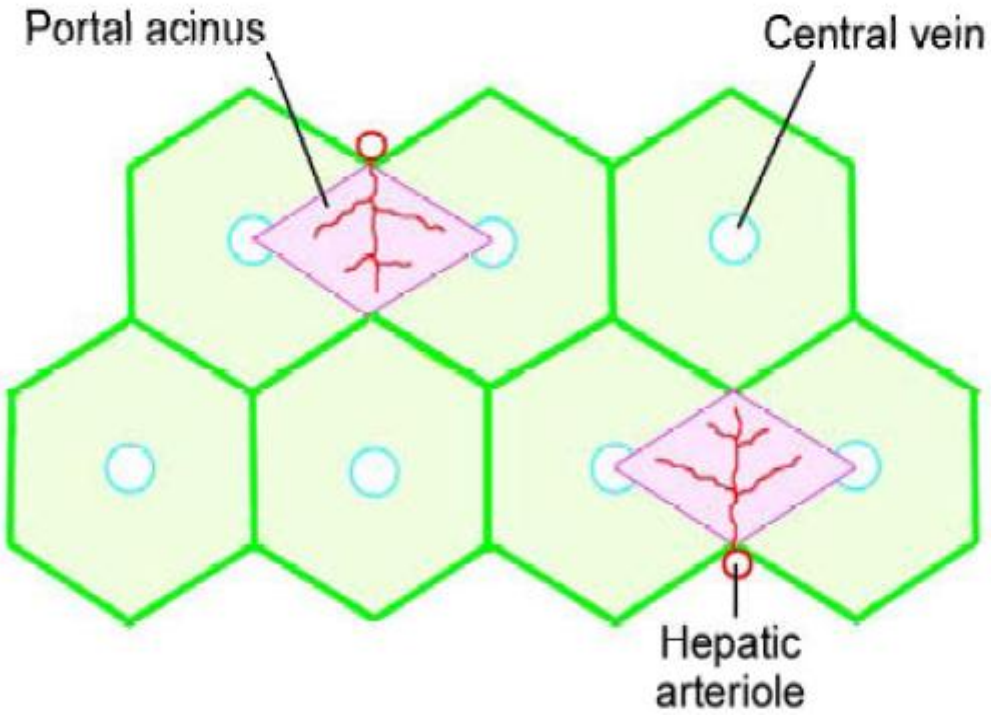


Portal lobule

- Triangular in shape.
- Boundary runs from one central vein to another.
- Axis is the bile duct in the portal triad.
- It represents the territory drained by an interlobular bile duct.

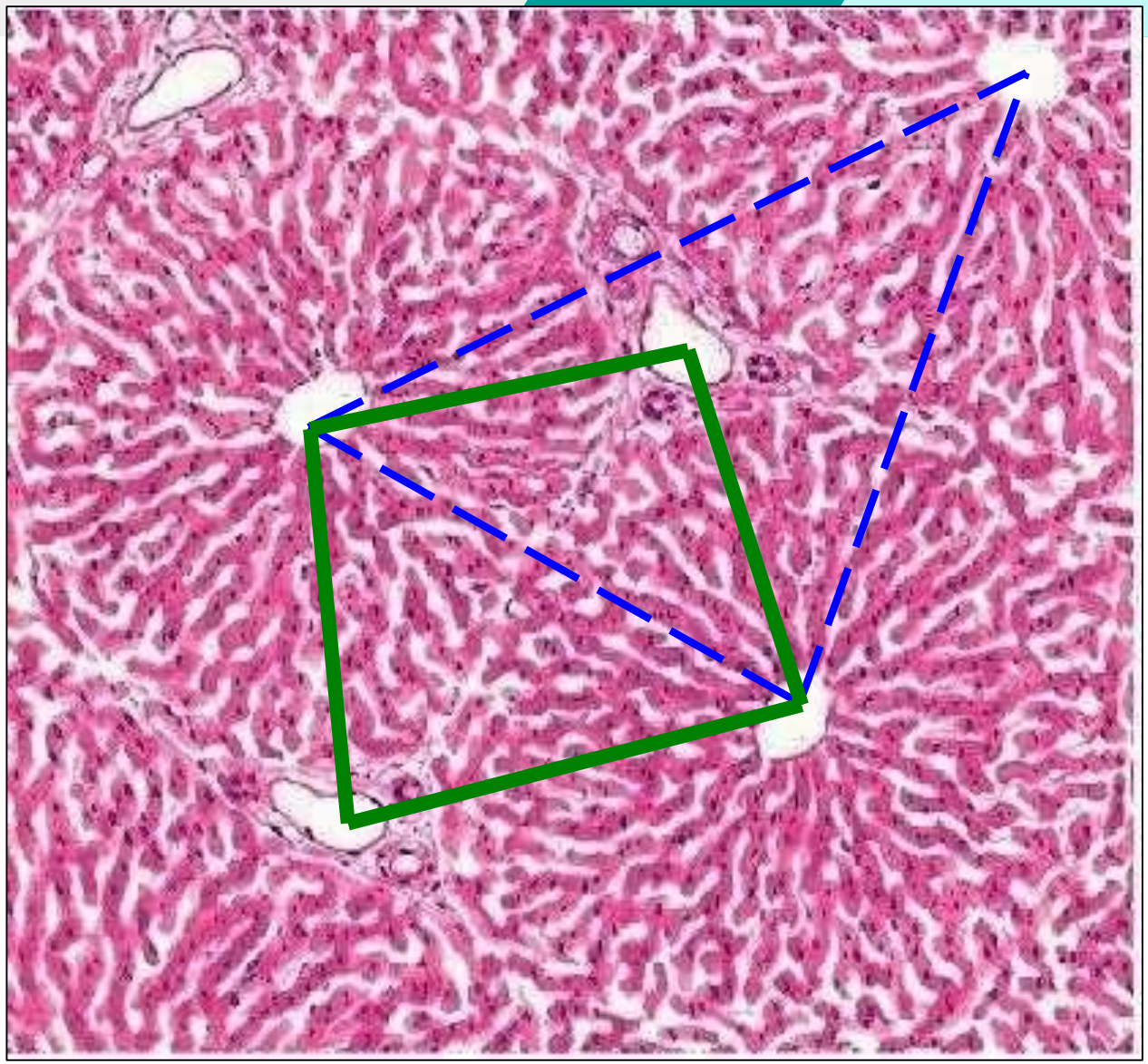


Portal Acinus



Hepatic acinus

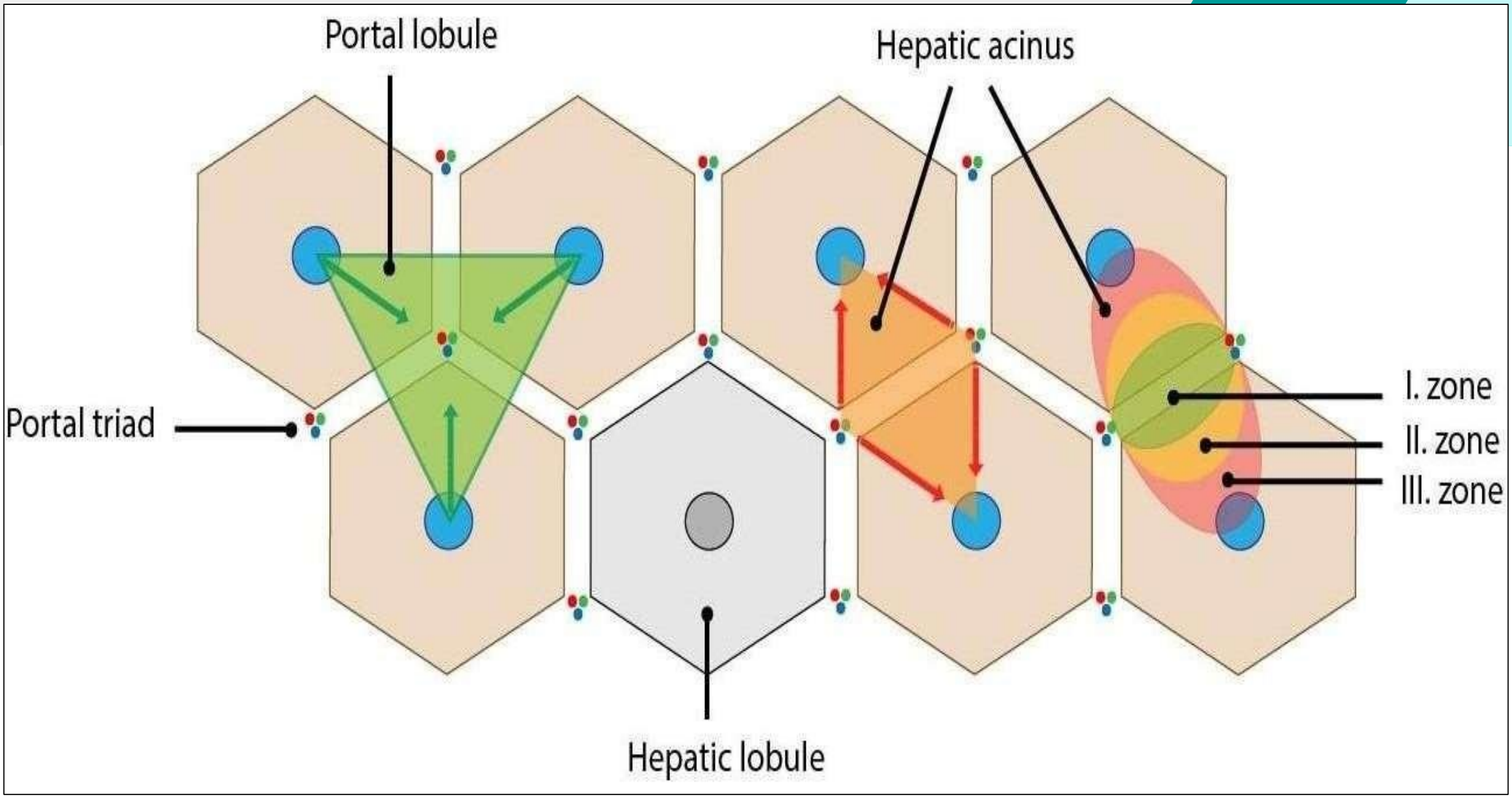
- Diamond shaped.
- Situated in adjacent areas of two hepatic lobules.
- Has portal triads at two ends and 2 central veins at facing ends.



- Hepatic acinus is subdivided into (3) zones.
  - Zone I: nearest to and between two portal triads.(best blood supply).
  - Zone II: next to zone I.
  - Zone III: nearest to central veins. (poorest blood supply, susceptible to damage).

- Cells of zone I are the 1st to be affected by or to alter by incoming blood.

  Eg.: in the event of fasting, cells in zone I are the first to respond to glucose-poor blood by breaking down glycogen and release glucose.


Portal lobule
Hepatic acinus
Portal triad
I. zone
II. zone
III. zone
Hepatic lobule

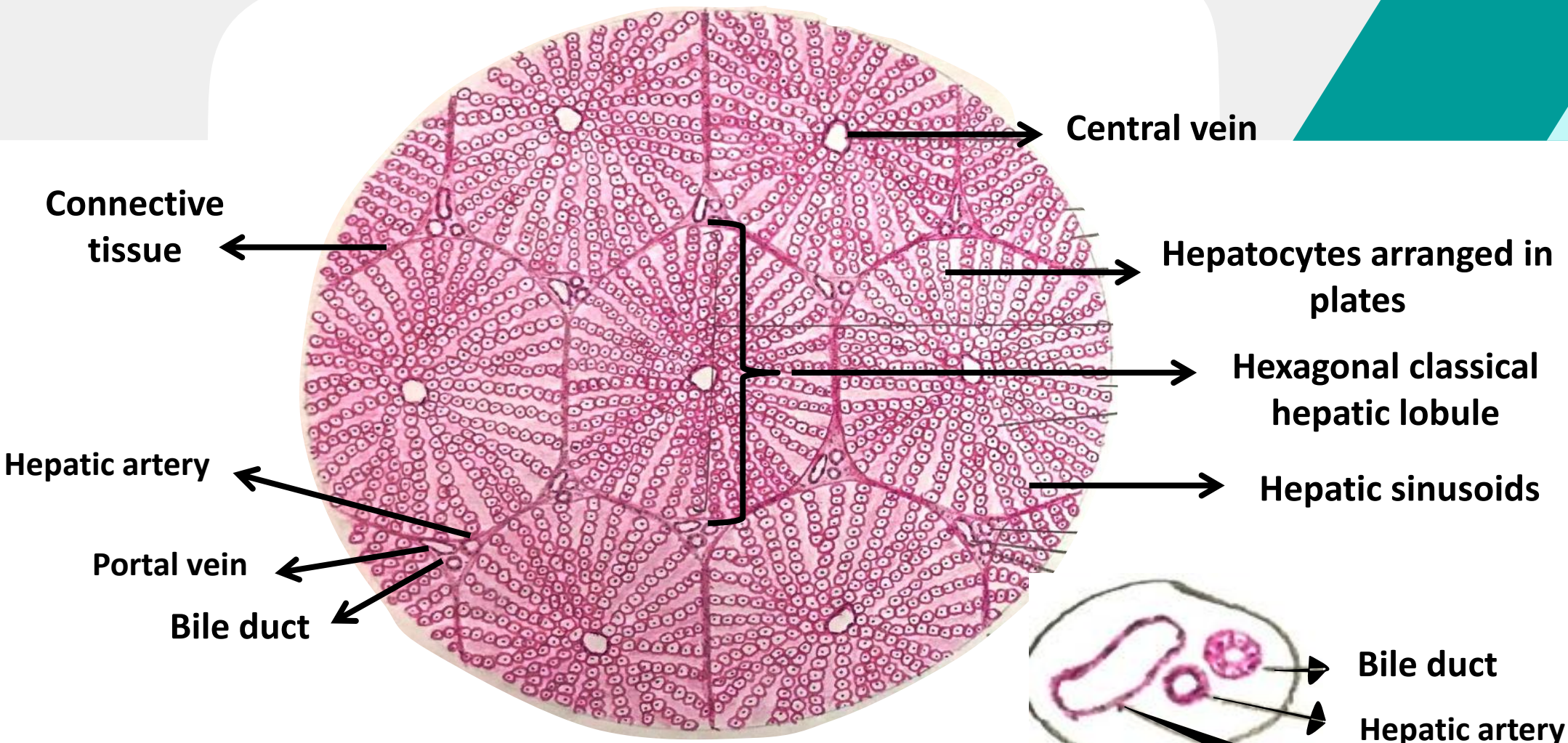


Fig. 10a. Section of liver

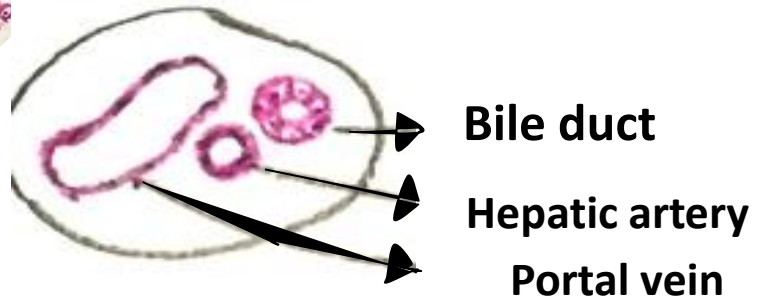
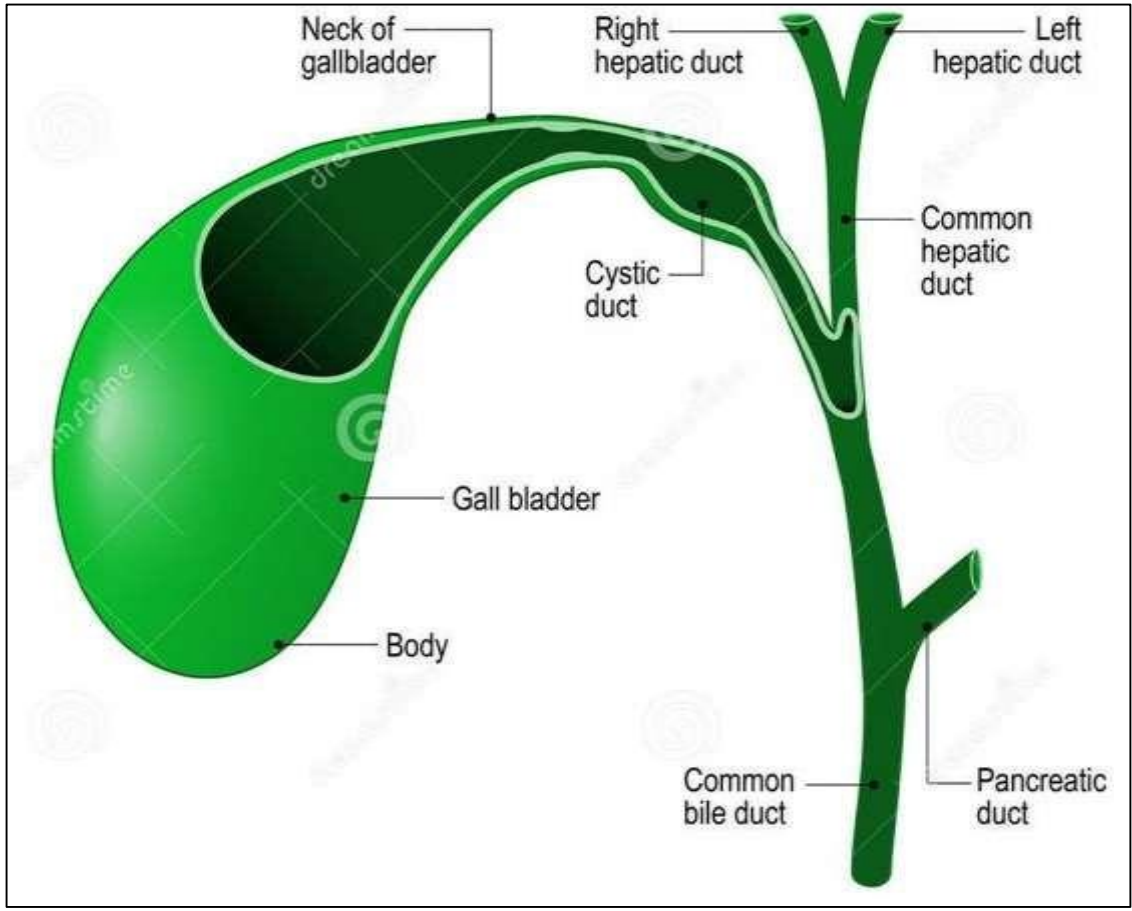


Fig. 10b. Portal triad

Gall bladder

- A piriform sac lies in a fossa in the right hepatic lobe's inferior surface.
- Upper surface- attached to the liver by connective tissue
- Completely covered by peritoneum- except superiorly.
- Parts- fundus, body and neck

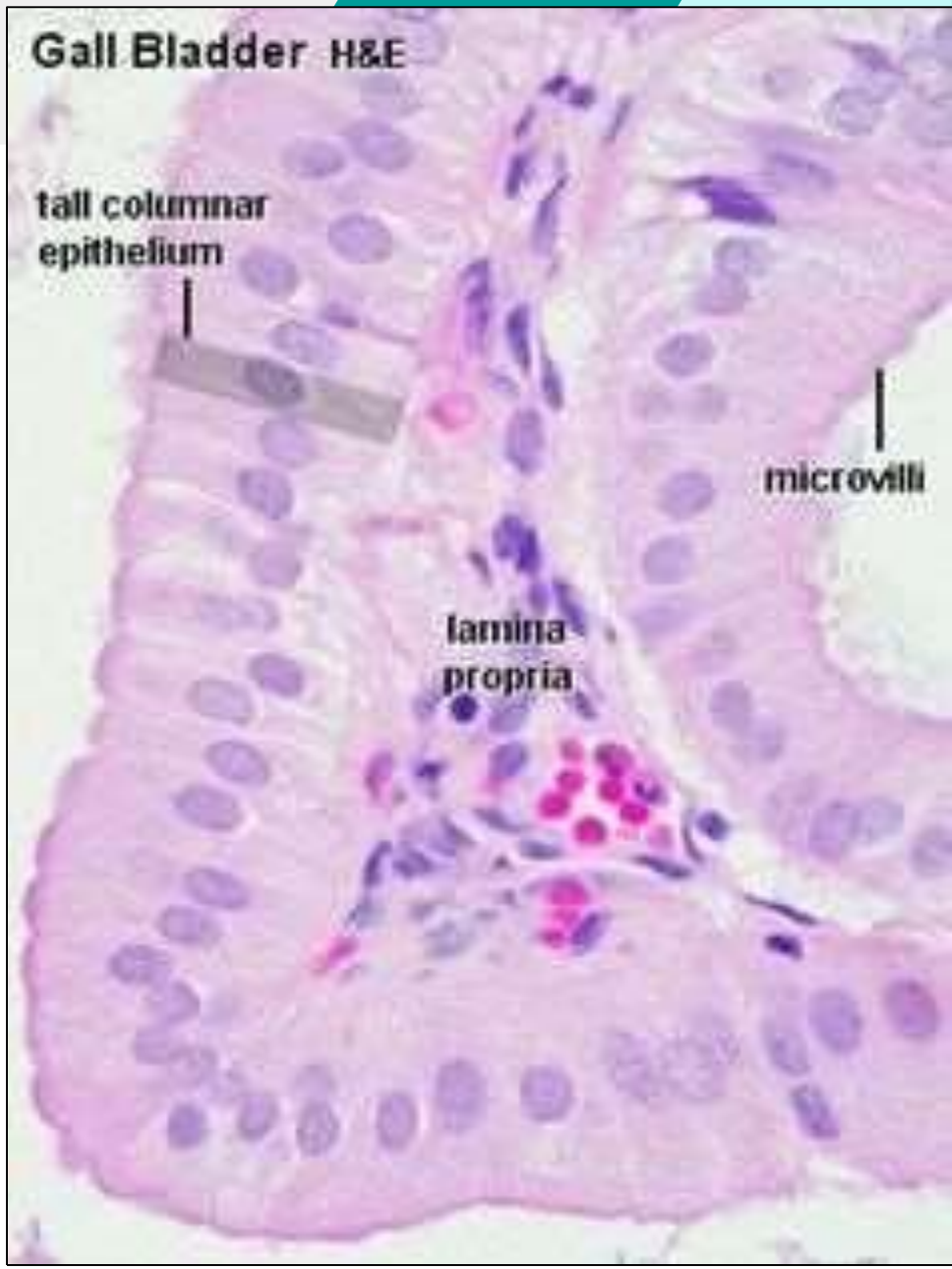


Structure

I. Mucosa

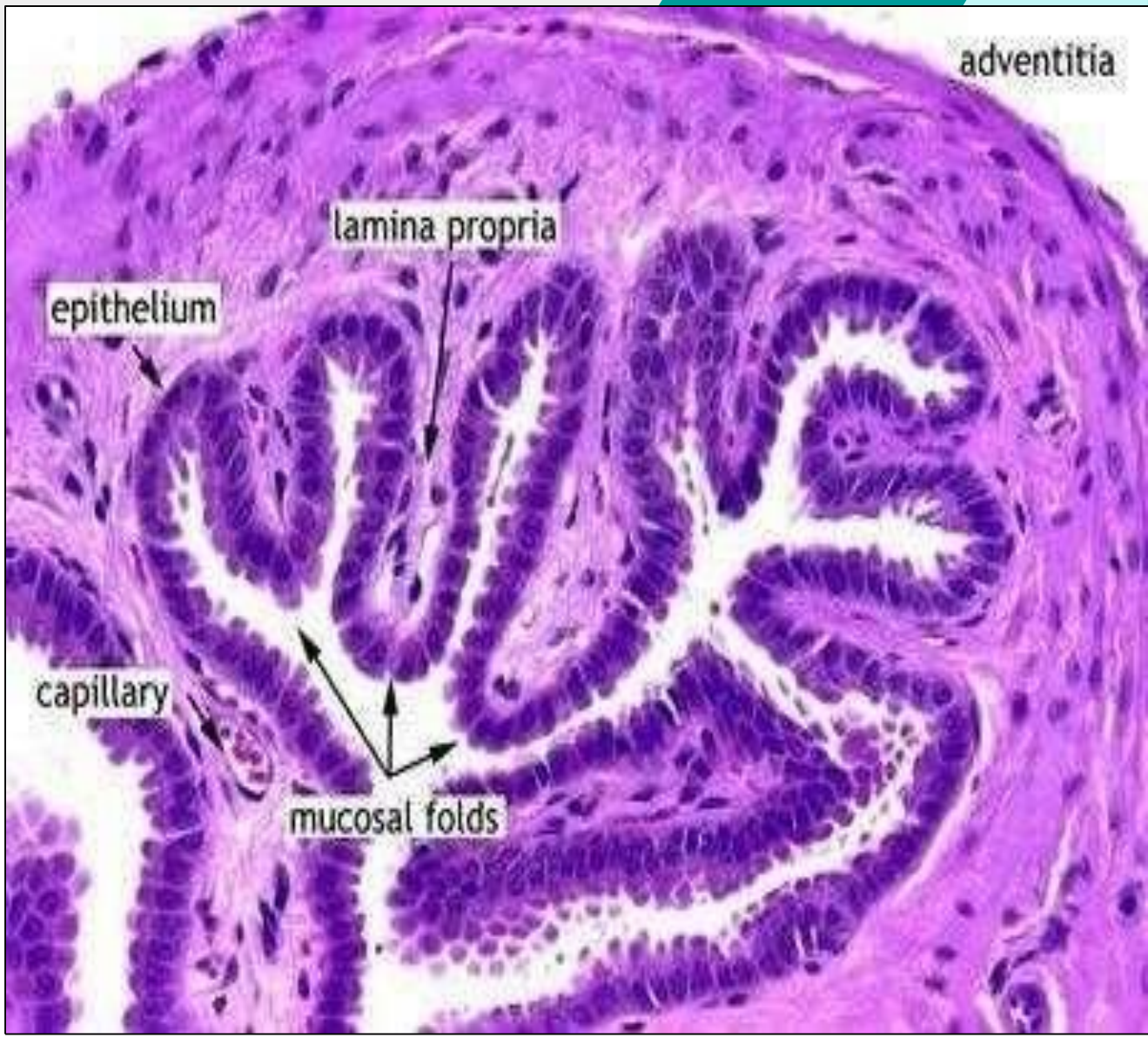
- **Epithelium** is simple columnar epithelium with irregular microvilli.

- **Lamina propria** contains loose connective tissue, some diffuse lymphatic tissue and blood vessels.

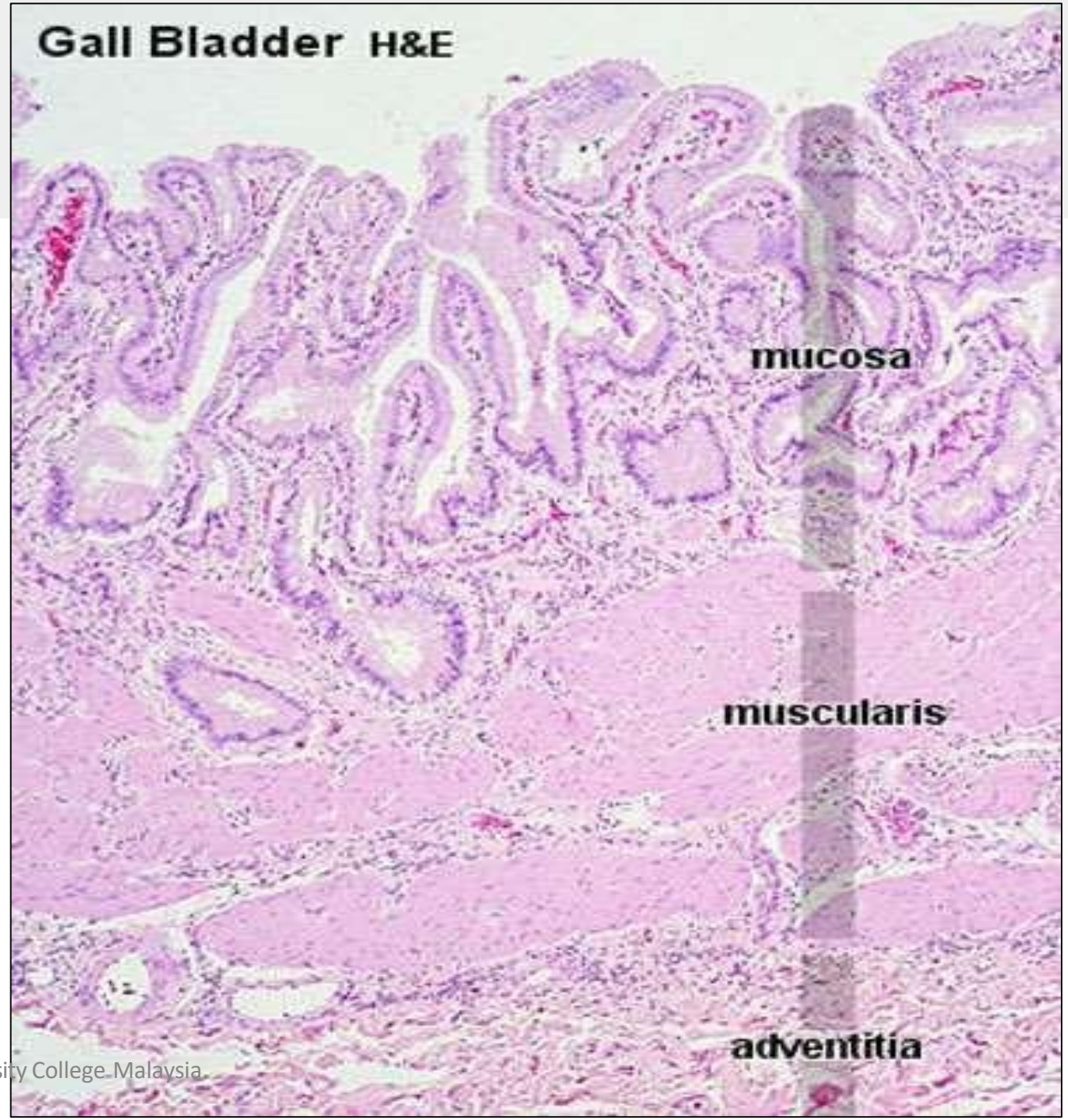


- Loose C.T. rich in elastic fibres, blood vessels and lymphatics which drain water reabsorbed from bile during the concentration process.

- In the neck region of gall bladder, mucous glands are found in the connective tissue layer.



In non-distended state, the gall bladder wall shows temporary mucosal folds, that disappear when it is distended with bile.



Submucosa is absent in gall bladder !

II. Muscularis

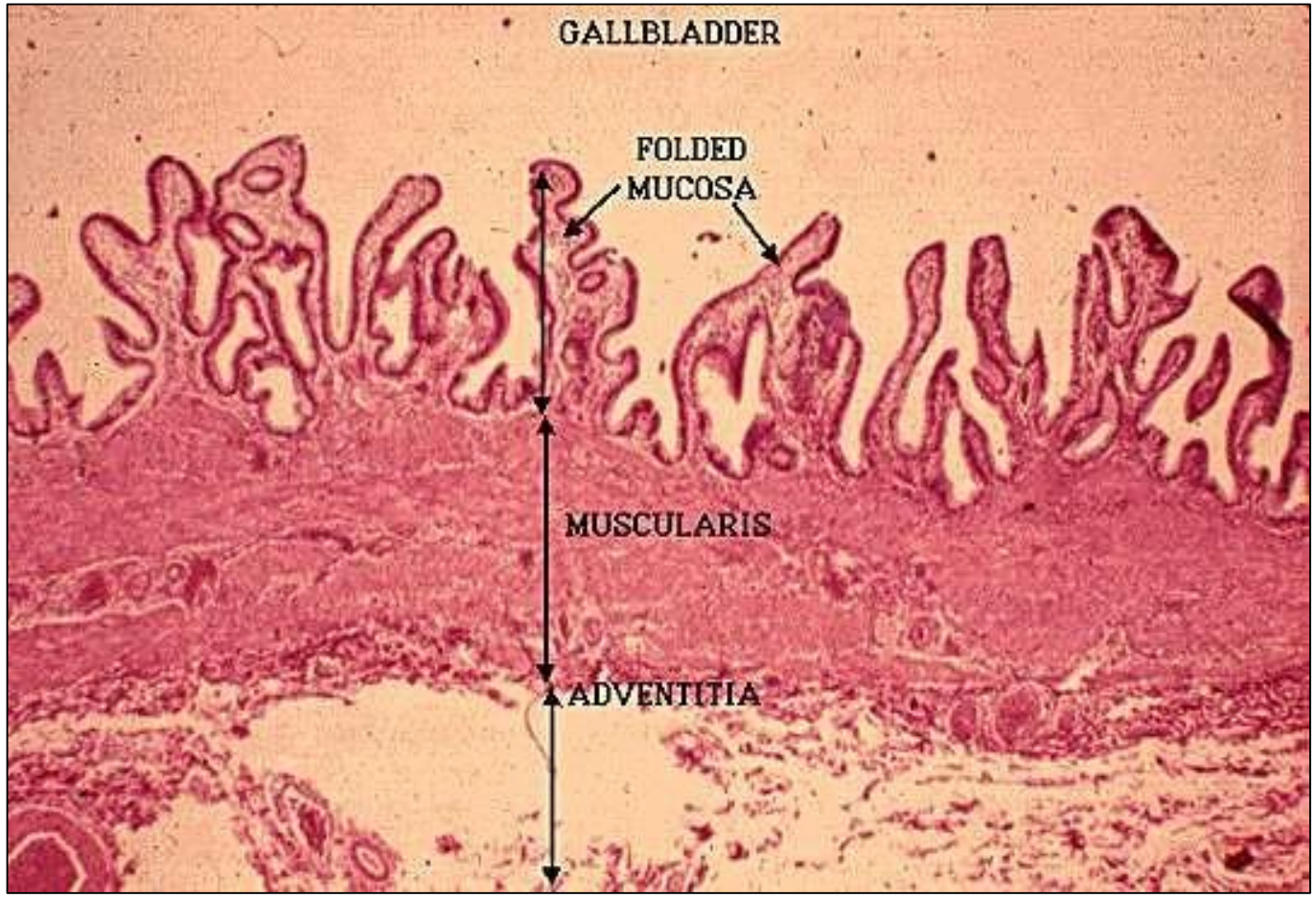
Composed of bundles of smooth muscle fibres arranged in longitudinal, transverse and oblique orientations but do not form distinct layers and interlacing elastic fibres.



III. Adventitia/ Serosa

- Dense connective tissue composed of collagen fibres and larger blood and lymphatic vessels.

- The surface attached to the liver is covered by adventitia and unattached surface is covered by serosa.



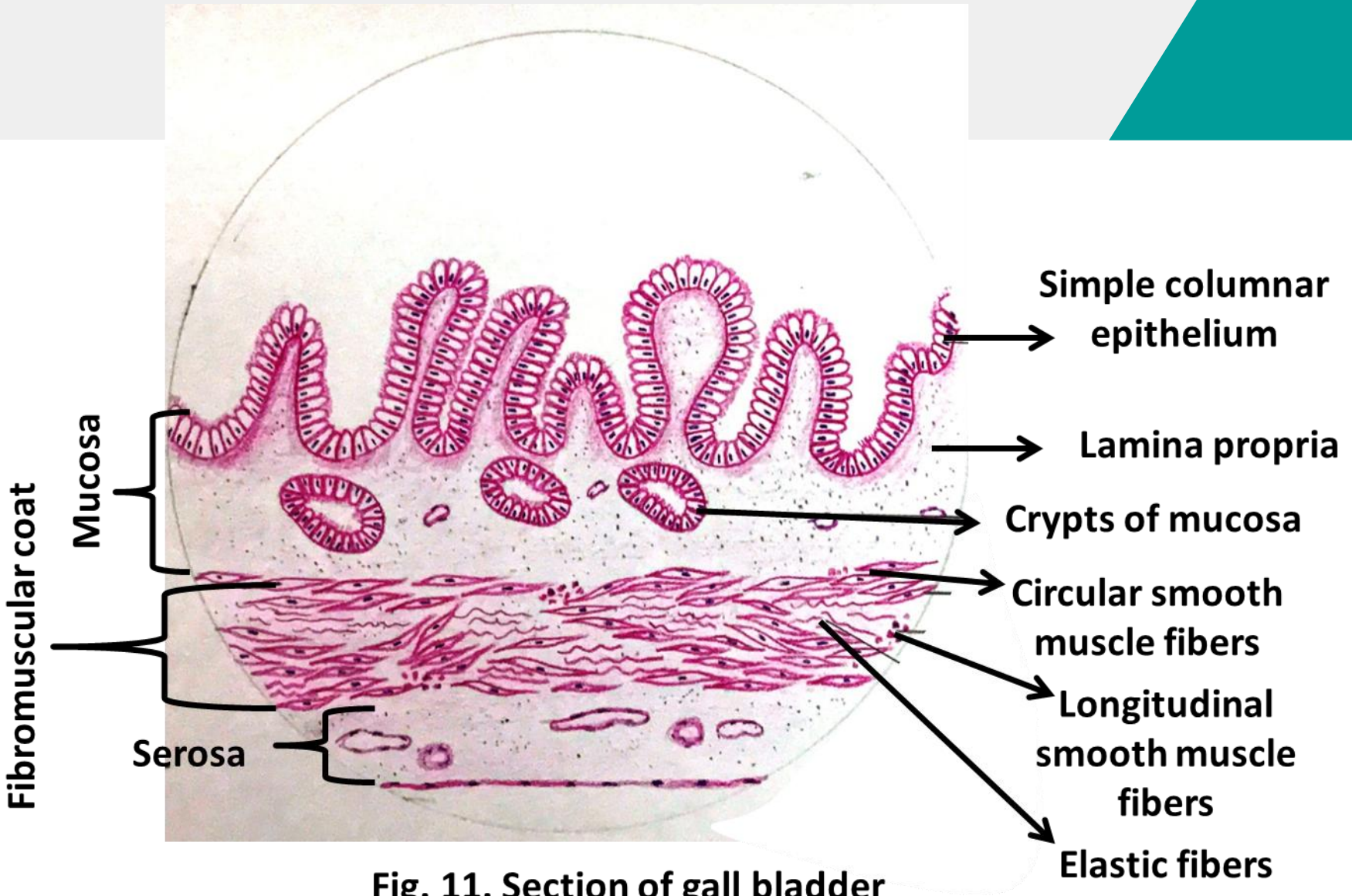
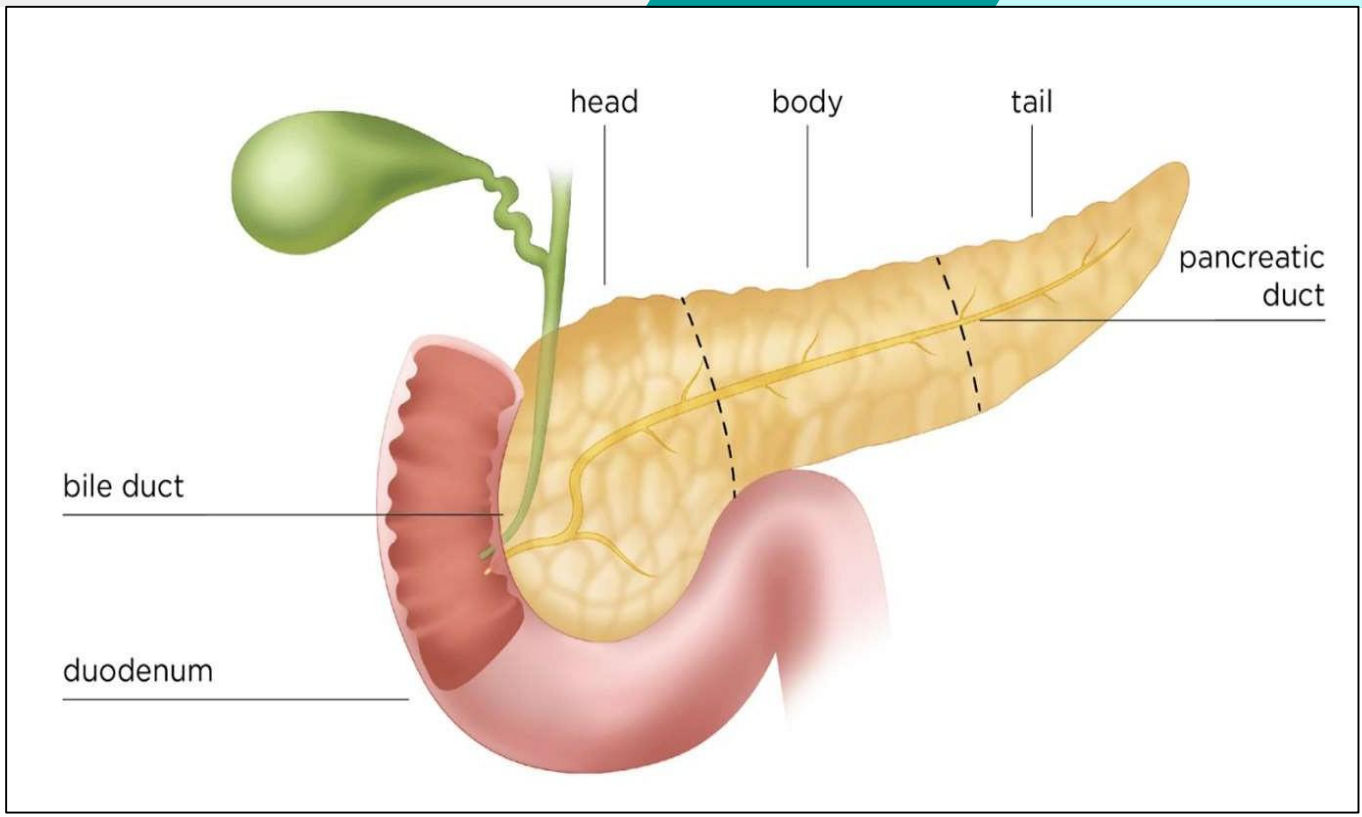


Fig. 11. Section of gall bladder

Pancreas

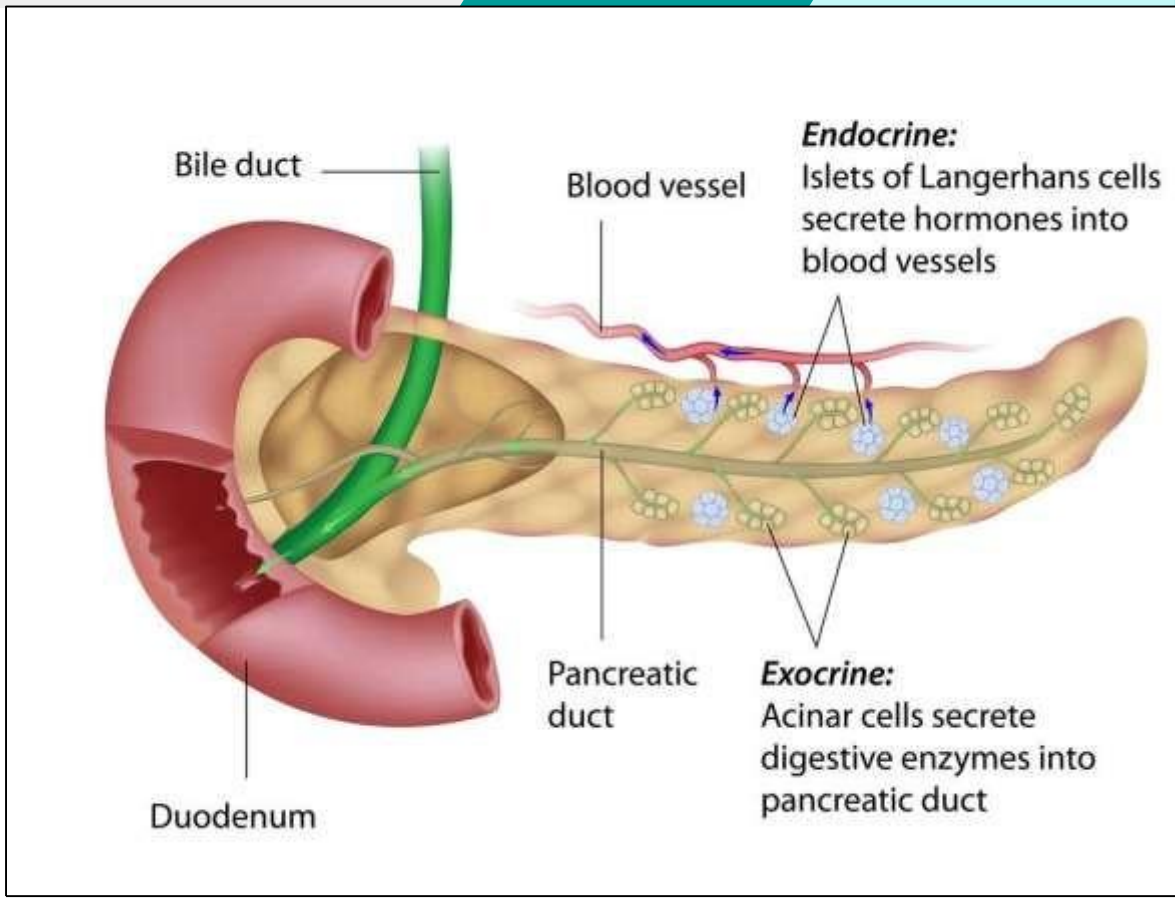
- Both an exocrine and an endocrine gland.
- The bulk of the gland is made up of acini which produce exocrine secretion which in turn is collected and delivered into the duodenum by duct system.



- Function:
  - ✓ Exocrine: enzymes
  - ✓ Endocrine: hormones

- Endocrine secretion is made by little clumps (islets of cells) richly supplied with capillaries scattered throughout the gland.

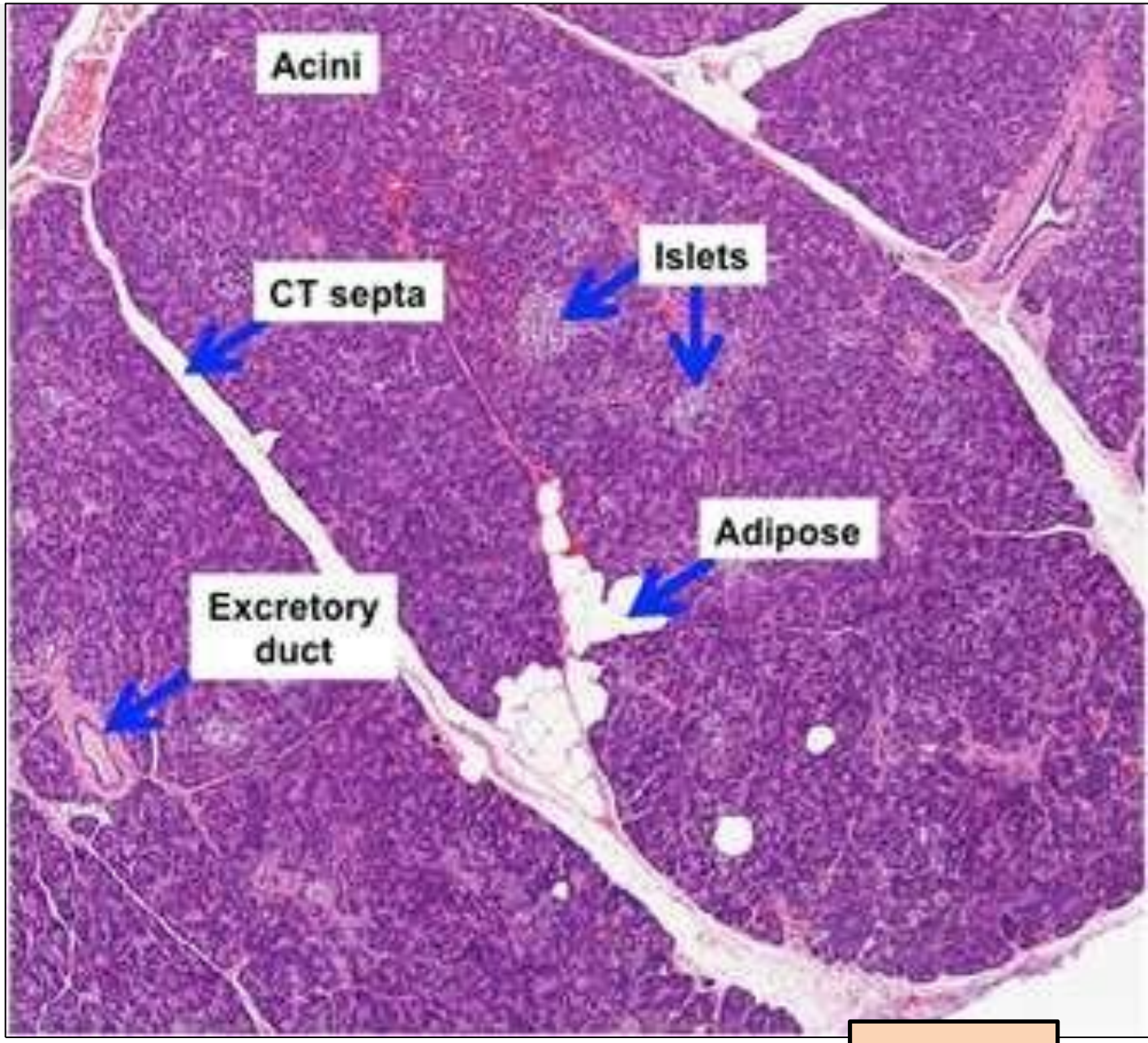
- These islets of Langerhans produce hormones like insulin and glucagon.



Structure

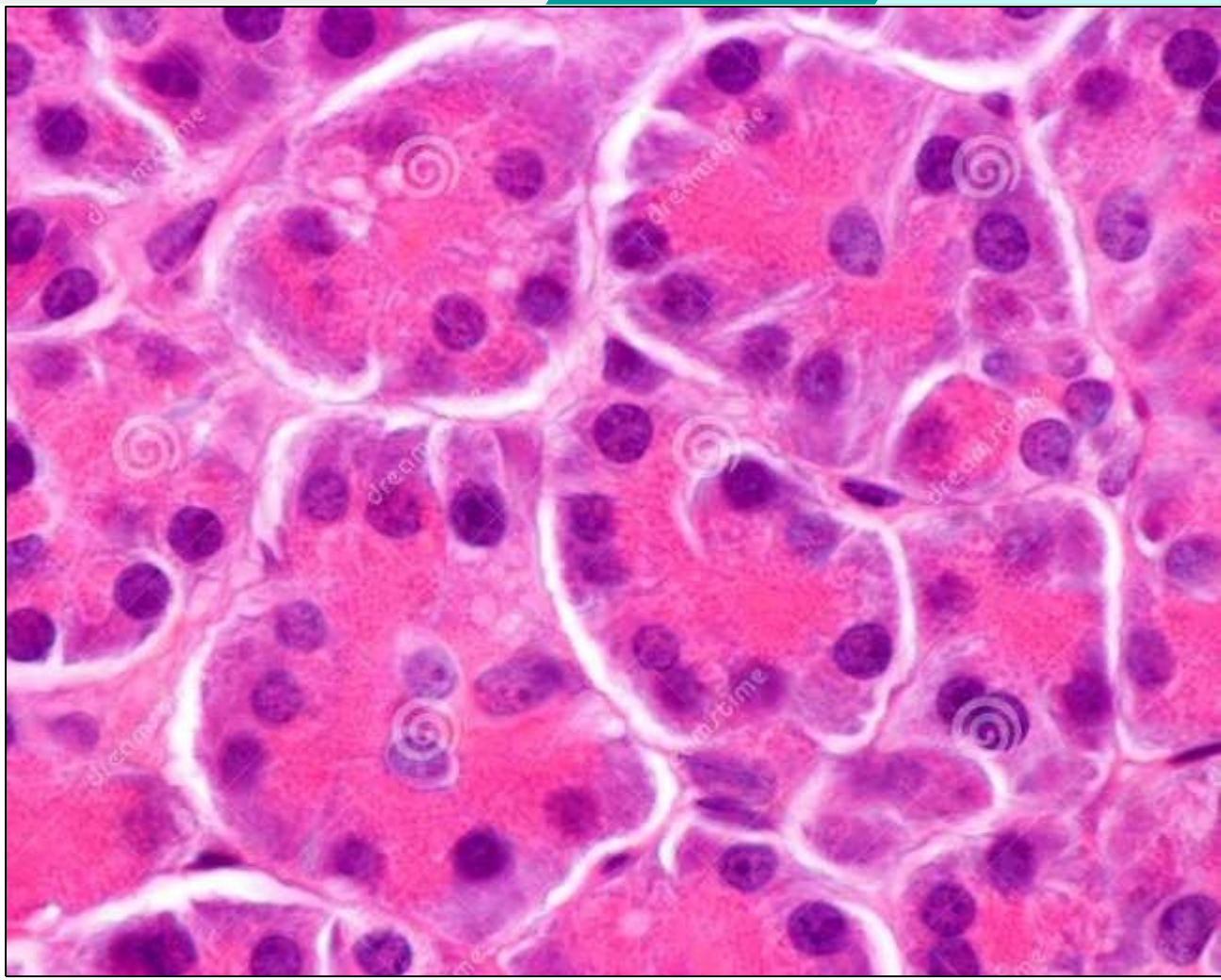
- Covered by a very thin connective tissue capsule which sends in septa within the gland and divides the pancreas into lobules.

- Most of the substance of lobules is made of exocrine pancreas with scattered endocrine islets.



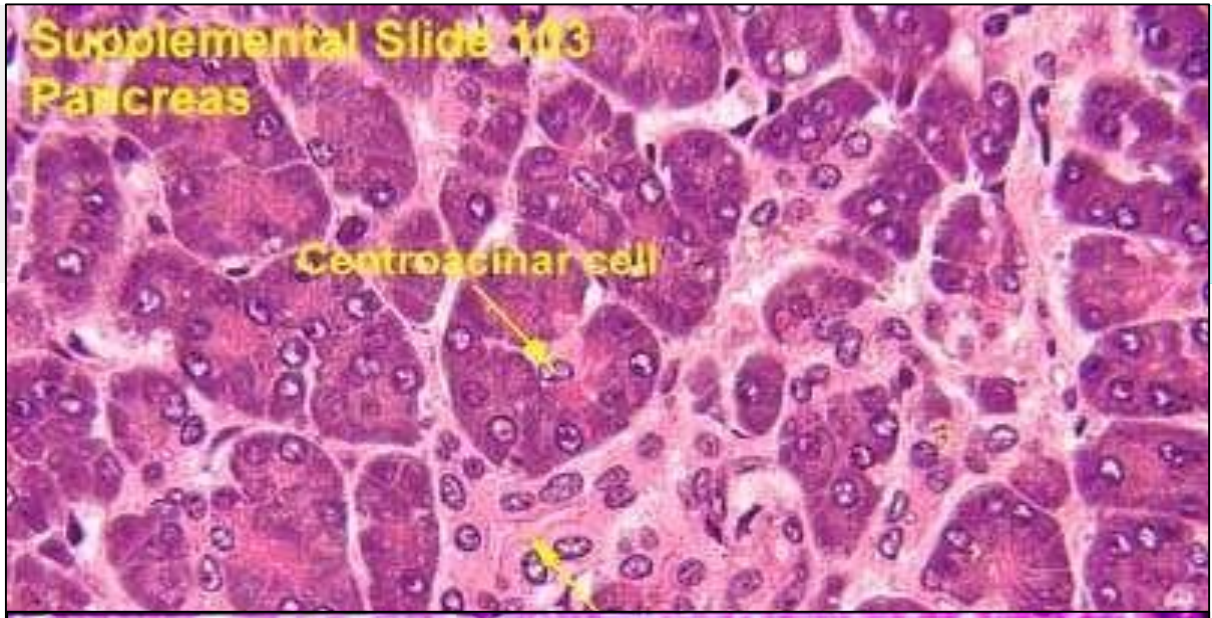
Exocrine pancreas

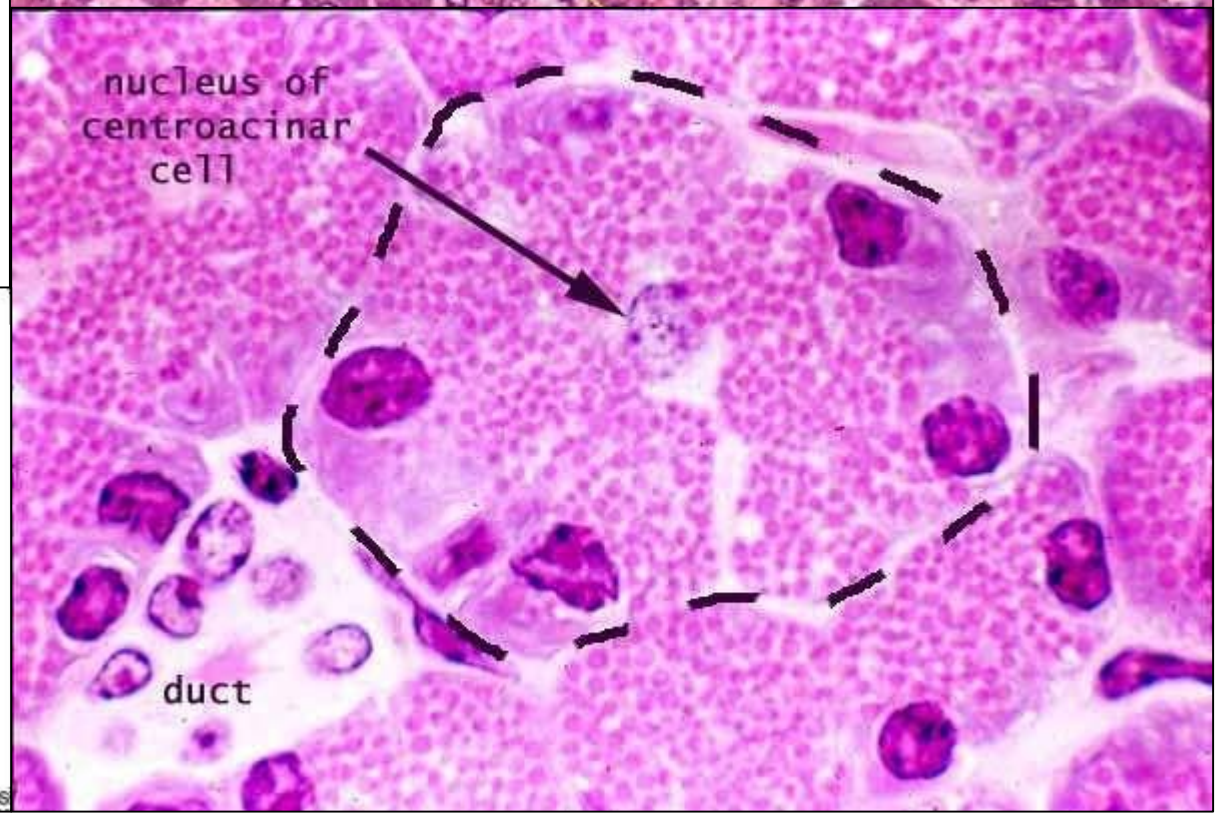
- Made up of acini and intralobular and interlobular ducts.
- Acini are typical serous acini. They are small rounded cluster of pyramidal cells with granules at the apex.
- Each cell has a rounded basal nucleus and basophilic cytoplasm.

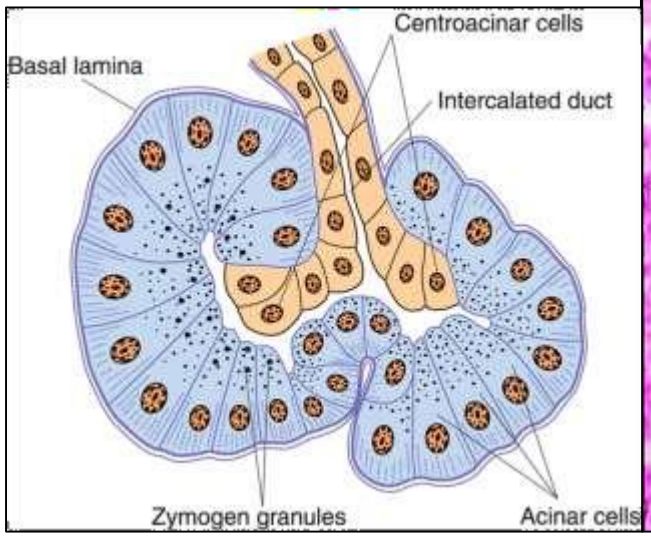


- The lumen shows the presence of centroacinar cells.

- Initial portion of the intercalated duct that extends up into the acinus.

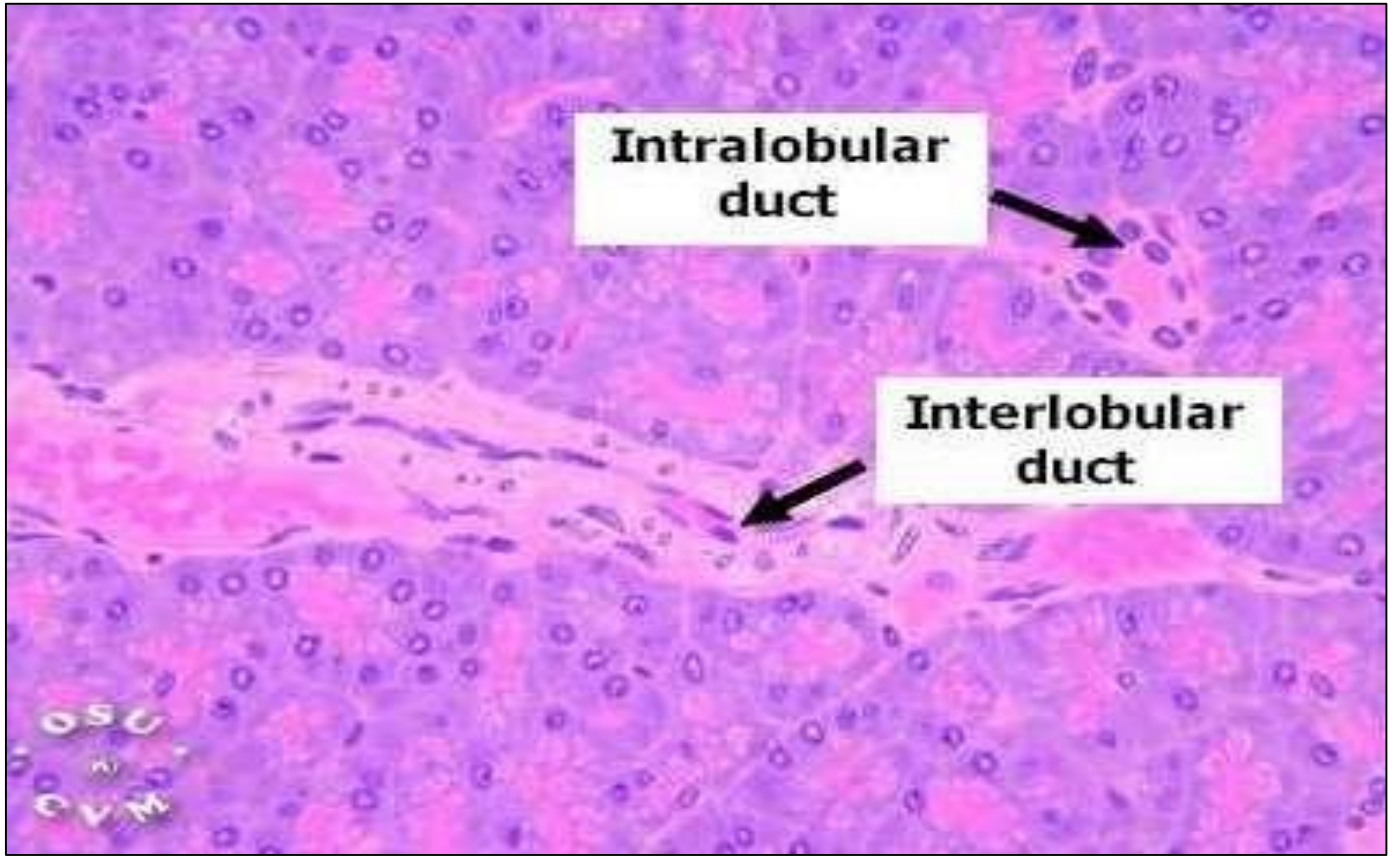






- Duct system:
  - ✓ Intercalated- cuboidal
  - ✓ Interlobular- columnar

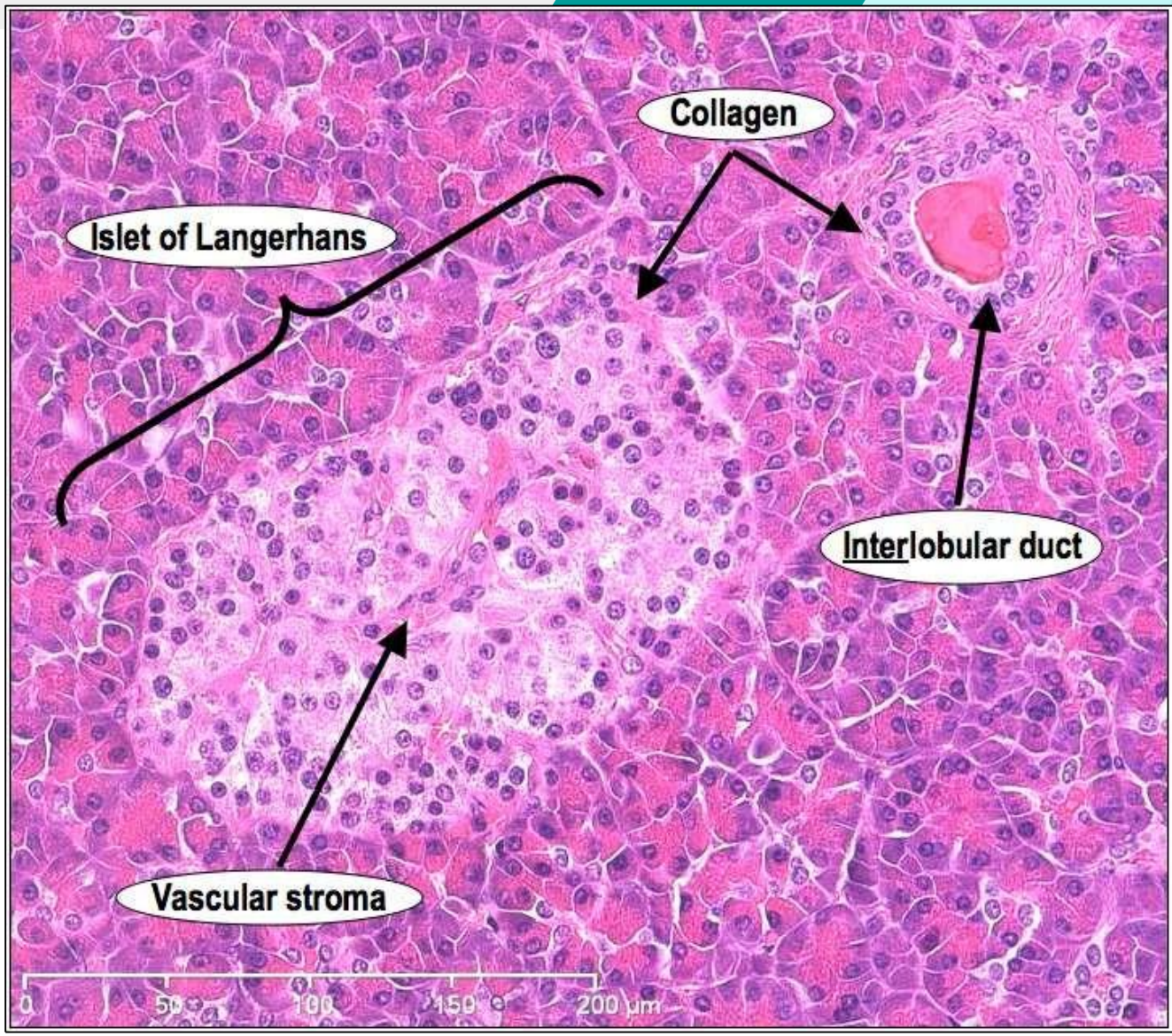
- Enzymes secreted by the acinar cells include proteases (trypsin), pancreatic amylase, lipase, and nuclease



Endocrine pancreas

Islets of Langerhans

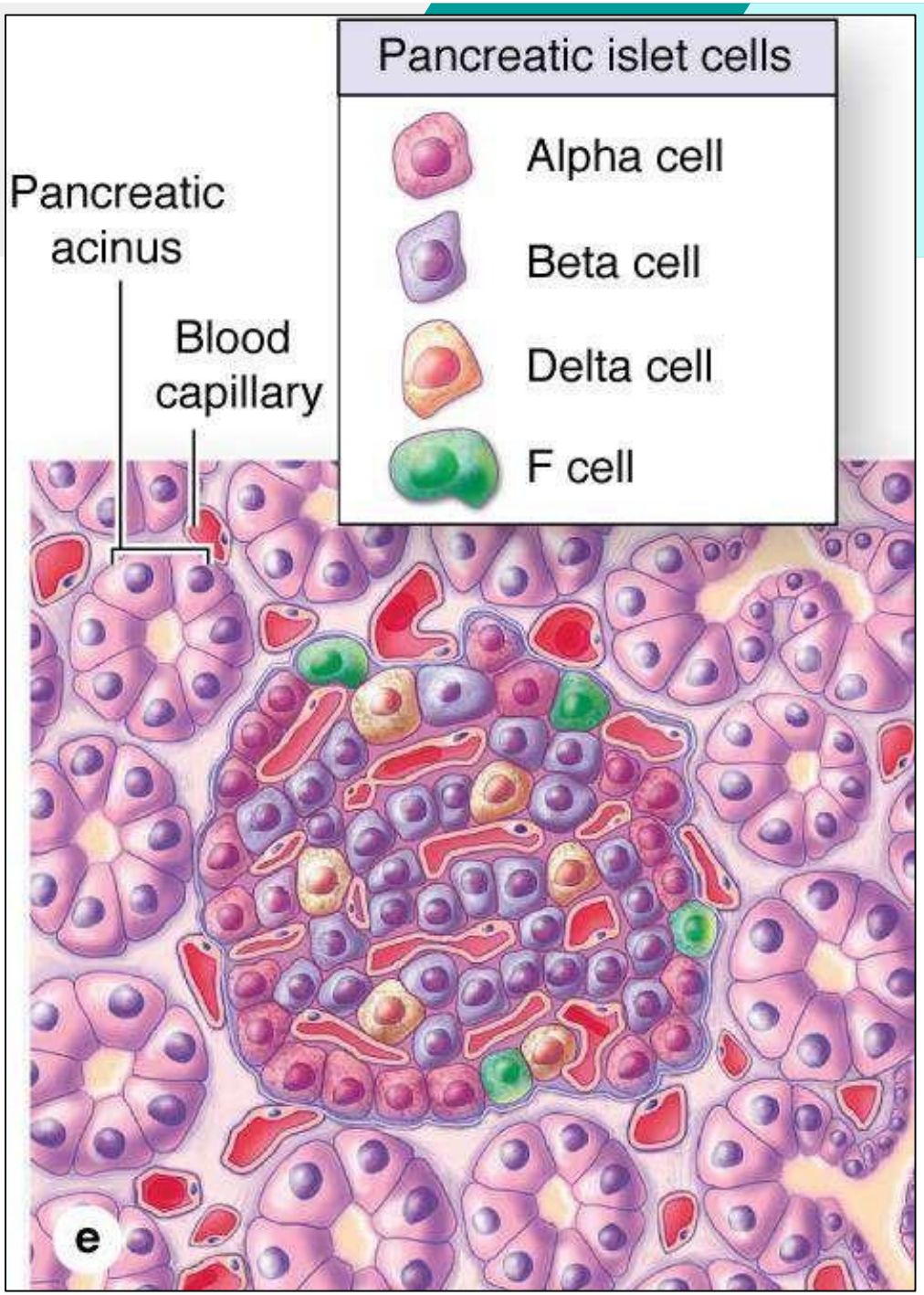
- Spheroidal/ polyhedral cluster of cells
- Most numerous in tail
- Pervaded by blood capillaries



3 types of Islet cells

Alpha, Beta, and Delta

- A cells -secretes glucagon
- B cells -secrete insulin
- D cells -secrete somatostatin



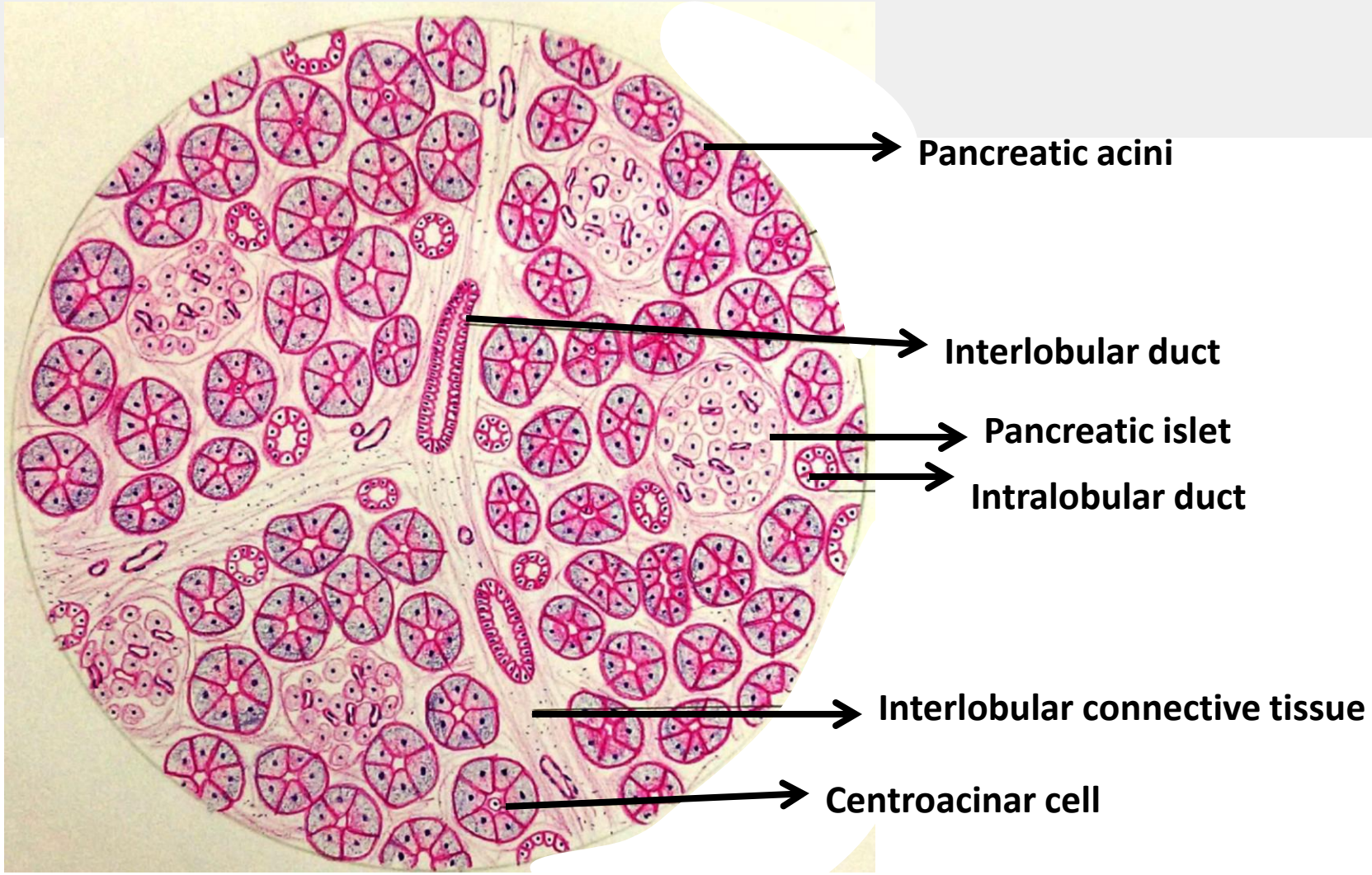


Fig. 4. Section of pancreas

THANK YOU !